

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
COLORADO RIVER BASIN REGION**

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**ORDER R7-2015-0030
NPDES NO. CA0104400**

**WASTE DISCHARGE REQUIREMENTS
FOR THE CITY OF IMPERIAL,
CITY OF IMPERIAL WATER POLLUTION CONTROL PLANT**

The following Discharger is subject to waste discharge requirements (WDRs) set forth in this Order:

Table 1. Discharger Information

Discharger	City of Imperial
Name of Facility	City of Imperial Water Pollution Control Plant
Facility Address	701 East 14 th Street
	Imperial, CA 92251
	Imperial County

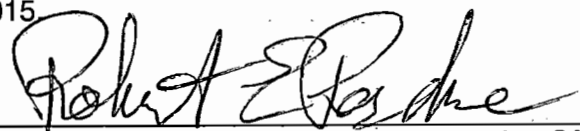
Table 2. Discharge Location

Discharge Point	Effluent Description	Discharge Point Latitude (North)	Discharge Point Longitude (West)	Receiving Water
001	Treated Municipal Wastewater	32° 51' 18" N	115° 33' 31" W	Dolson Drain

Table 3. Administrative Information

This Order was adopted on:	September 17, 2015
This Order shall become effective on:	October 1, 2015
This Order shall expire on:	September 30, 2020
The Discharger shall file a Report of Waste Discharge as an application for reissuance of WDRs in accordance with title 23, California Code of Regulations, and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than:	April 3, 2020, or as soon as possible if planned changes meet the Notice Requirement under 40 C.F.R. 122.41(l) (1).
The U.S. Environmental Protection Agency (U.S. EPA) and the California Regional Water Quality Control Board, Colorado River Basin Region have classified this discharge as follows:	Major Discharge

I, Robert Perdue, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Colorado River Basin Region, on September 17, 2015.



Robert Perdue, Executive Officer

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I. FACILITY INFORMATION

The following Discharger is subject to WDRs as set forth in this Order:

Table 4. Facility Information

WDID	7A130106011
Discharger	City of Imperial
Name of Facility	City of Imperial Water Pollution Control Plant
Facility Address	701 East 14 th Street
	Imperial, CA 92251
	Imperial County
Legally Responsible Official	Jackie Loper, Public Services Director (760) 355-3336; jloper@cityofimperial.org
Facility Contact, Title and Phone	Mark Coon, Chief Wastewater Operator (760) 455-3426; mark1605ww@hotmail.com
Authorized Person(s) to Sign and Submit Reports	Mark Coon, Chief Wastewater Operator (760) 455-3426; mark1605ww@hotmail.com Edgar Self, Public Service Manager (760) 355-2734; eself@cityofimperial.org
Mailing Address	420 South Imperial Avenue Imperial, CA 92251
Billing Address	Same
Type of Facility	Publicly Owned Treatment Works (POTW)
Major or Minor Facility	Major
Threat to Water Quality	2
Complexity	A
Pretreatment Program	N
Reclamation Requirements	N
Facility Permitted Flow	2.4 million gallons per day (MGD)
Facility Design Flow	2.4 MGD
Watershed	West Colorado River Basin
Receiving Water	Dolson Drain, a tributary to Alamo River via Lilac Drain and Rose Drain
Receiving Water Type	Agricultural Drain

II. FINDINGS

The California Regional Water Quality Control Board, Colorado River Basin Region (Colorado River Basin Water Board), finds:

- A. Background.** The City of Imperial (hereinafter Discharger) is currently discharging pursuant to Order R7-2010-0020, as amended by Special Order R7-2011-0018, and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0104400. The Discharger submitted a Report of Waste Discharge (ROWD) dated April 3, 2015, and applied for a NPDES permit renewal discharge of up to 2.4 MGD of treated wastewater from the City of Imperial Water Pollution Control Plant, hereinafter Facility. The application was deemed complete on June 4, 2015. A site visit was conducted on July 28, 2015, to observe operations and collect additional data to develop permit limitations and conditions.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein. Additional definitions are provided in Attachment A.

- B. Facility Description.** The Discharger owns and operates a municipal wastewater treatment plant. The total design capacity of the wastewater treatment plant is 2.4 MGD. The current wastewater treatment plant consists of an influent pumping station, grit chamber, two parallel treatment trains (oxidation ditches followed by secondary clarification and extended aeration/activated sludge treatment via a Biolac® system), ultraviolet (UV) disinfection system, and sludge drying beds. Wastewater is discharged from Discharge Point 001 (see Table 2 on cover page) to the Dolson Drain, a water of the United States, tributary to the Alamo River via Lilac Drain and Rose Drain, within the Salton Sea Watershed. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.
- C. Legal Authorities.** This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters.
- D. Background and Rationale for Requirements.** The Colorado River Basin Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E and G are also incorporated into this Order.
- E. Provisions and Requirements Implementing State Law.** The provisions/requirements in subsections IV.B, IV.C, V.B, and VI.C.4 of this Order are included to implement state law only. These provisions/requirements are not required or authorized under the federal CWA (33 U.S.C. § 1251 et seq.); consequently, violations of these provisions/requirements are not subject to the enforcement remedies that are available for NPDES violations.
- F. Notification of Interested Parties.** The Colorado River Basin Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet.

G. Consideration of Public Comment. The Colorado River Basin Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED that Board Order R7-2010-0020, as amended by Special Order R7-2011-0018, is rescinded upon the effective date of this Order except for enforcement purposes, and, in order to meet the provisions contained in division 7 of the Water Code (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations adopted thereunder, the Discharger shall comply with the requirements in this Order. This action in no way prevents the Colorado River Basin Water Board from taking enforcement action for past violations of the previous Orders.

III. DISCHARGE PROHIBITIONS

- A.** The discharge of waste to land not owned or controlled by the Discharger is prohibited unless authorized in Waste Discharge Requirements or NPDES Permit.
- B.** Discharge of treated wastewater at a location or in a manner different from that described in Findings of this Order is prohibited.
- C.** The discharge of trash to the Dolson Drain is prohibited.
- D.** Except as allowed under the Standard Provisions for NPDES permits (hereinafter Standard Provisions), included as Attachment D, the bypass or overflow of untreated wastewater or wastes to the Dolson Drain is prohibited.
- E.** The Discharger shall not accept waste in excess of the design treatment or disposal capacity of the system.
- F.** The discharge shall not cause degradation of any water supply.
- G.** The treatment or disposal of wastes from the facility shall not cause pollution or nuisance as defined in Section 13050, subdivisions (l) and (m), respectively, of the California Water Code.

IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Effluent Limitations – Discharge Point 001

1. Final Effluent Limitations – Discharge Point 001

- a. The Discharger shall maintain compliance with the following effluent limitations at Discharge Point 001, with compliance measured at Monitoring Location EFF-001 as described in the Monitoring and Reporting Program (MRP), Attachment E:

Table 5. Effluent Limitations

Parameter	Units	Effluent Limitations				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Flow	MGD	2.4	---	---	---	---
pH	Standard Units	---	---	---	6.0	9.0
Biochemical Oxygen Demand 5-day @ 20°C	mg/L	30	45	---	---	---
	lbs/day ¹	601	901	---	---	---
Suspended Solids, Total	mg/L	30	45	---	---	---
	lbs/day ¹	601	901			
Oil and Grease, Total	mg/L	---	---	25	---	---
	lbs/day ¹	---	---	500	---	---

Parameter	Units	Effluent Limitations				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Copper, Total Recoverable	µg/L	9.8	---	30.4	---	---
	lbs/day ¹	0.19	---	0.61	---	---

¹ The mass-based effluent limitations are based on a design capacity of 2.4 MGD.

- b. **Percent Removal:** The average monthly percent removal of BOD 5-day 20°C and total suspended solids (TSS) shall not be less than 85 percent.
- c. **Toxicity:** There shall be no toxicity in the treatment plant effluent nor shall the treatment plant effluent cause any toxicity in the receiving water, as defined in section V of Attachment E, MRP. All waters shall be maintained free of toxic substances in concentrations which are toxic to, or which produce detrimental physiological responses in human, plant, animal, or indigenous aquatic life. Compliance with this objective will be determined by use of indicator organisms, analyses of species diversity, population density, growth anomalies, or toxicity tests of appropriate duration or other appropriate methods specified by the Colorado River Basin Water Board.
- d. **Bacteria:** The bacterial density in the wastewater effluent discharged to the Dolson Drain, shall not exceed the following values, as measured by the following bacterial indicators:
 - i. ***E. coli*.** The geometric mean bacterial density (based on a minimum of not less than five samples equally spaced over a 30-day period) shall not exceed a Most Probable Number (MPN) of 126 per 100 milliliters, nor shall any sample exceed the maximum allowable bacterial density of a MPN of 400 per 100 milliliters.
 - ii. **Fecal Coliform.** The geometric mean bacterial density (based on a minimum of not less than five samples equally spaced over a 30-day period) shall not exceed a MPN of 200 per 100 milliliters, nor shall more than ten percent of the total samples during any 30-day period exceed a MPN of 400 per 100 milliliters.
- e. **Total Dissolved Solids:** Discharges of wastes or wastewater shall not increase the total dissolved solids content of receiving waters, unless it can be demonstrated to the satisfaction of the Colorado River Basin Water Board that such an increase in total dissolved solids does not adversely affect beneficial uses of receiving waters.

2. Interim Effluent Limitations – Not Applicable

B. Land Discharge Specifications – Not Applicable

C. Recycling Specifications – Not Applicable

V. RECEIVING WATER LIMITATIONS

A. Surface Water Limitations

Receiving water limitations are based on water quality objectives contained in the Basin Plan and are a required part of this Order. The discharge shall not cause the following in the Dolson Drain:

1. Result in the concentration of dissolved oxygen in the receiving water to fall below 5.0 mg/L. When dissolved oxygen in the receiving water is already below 5.0 mg/L, the discharge shall not cause any further depression.

2. Result in the presence of oil, grease, floating material (liquids, solids, foam and scum) or suspended material in amounts that create a nuisance or adversely affect beneficial uses.
3. Result in the deposition of pesticides or combination of pesticides detectable in concentrations that adversely affects beneficial uses.
4. Result in discoloration in the receiving water that adversely affects beneficial uses.
5. Result in the discharge of biostimulatory substances in concentrations that promote aquatic growths to the extent that such growths cause nuisance or adversely affect beneficial uses.
6. Result in an increase of turbidity that adversely affects beneficial uses.
7. Result in the normal ambient pH of the receiving water to fall below 6.0 or exceed 9.0 units.
8. Result in altering the natural receiving water temperature that adversely affects beneficial uses.
9. Result in the deposition of material that causes nuisance or adversely affects beneficial uses.
10. Result in the discharge of an individual chemical or combination of chemicals in concentrations that adversely affect beneficial uses.
11. Result in toxic pollutants to be present in the water column, sediments or biota in concentrations that adversely affect beneficial uses or that produce detrimental physiological responses in human, plant, animal, or aquatic life.
12. Result in an increase in taste or odor-producing substances that adversely affect beneficial uses.
13. Result in the violation of any applicable water quality standard for receiving waters adopted by the Colorado River Basin Water Board or the State Water Board as required by the Federal CWA and regulations adopted thereunder. If more stringent applicable water quality standards are promulgated or approved pursuant to CWA section 303 or amendments thereto, the Colorado River Basin Water Board will revise and modify this Permit in accordance with such more stringent standard.
14. Result in the concentration of total dissolved solids in the Dolson Drain to exceed an annual average concentration of 4,000 mg/L or a maximum daily concentration of 4,500 mg/L.

B. Groundwater Limitations – Not Applicable

VI. PROVISIONS

A. Standard Provisions

1. **Federal Standard Provisions.** The Discharger shall comply with all Standard Provisions included in Attachment D of this Order.
2. **Colorado River Basin Water Board Standard Provisions.** The Discharger shall comply with the following provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply:

- a. The POTW shall be protected from any washout or erosion of wastes or covering material, and from any inundation, which could occur as a result of floods having a predicted frequency of once in 100 years.
- b. The Discharger shall comply with all conditions of this Order. Noncompliance constitutes a violation of the Federal Clean Water Act and Porter-Cologne Water Quality Control Act, and is grounds for enforcement action; for permit termination, revocation and reissuance, or modification of waste discharge requirements; or denial of a permit renewal application.
- c. The Discharger shall ensure that all site-operating personnel are familiar with the contents of this Order, and shall maintain a copy of this Order at the site.
- d. The Discharger shall immediately notify the Office of Emergency Services by phone at (800) 852-7550 to report any noncompliance that may endanger human health or the environment as soon as: (1) the Discharger has knowledge of the discharge, (2) notification is possible, and (3) notification can be provided without substantially impeding cleanup or other emergency measures.

Although State and Regional Water Boards do not have duties as first responders, it is important to ensure that the agencies that do have first responder duties are notified in a timely manner in order to protect public health and beneficial uses. To carry out this objective, the following notification requirements are to be implemented:

- i. For any discharges of sewage that result in a discharge to a drainage channel or surface water, the Discharger shall, as soon as possible, but not later than two (2) hours after becoming aware of the discharge, notify the State Office of Emergency Services.
- ii. As soon as possible, follow the notification, reporting, monitoring, and recordkeeping requirements under WQ 2013-0058-EXEC for the Statewide Waste Discharge Requirements for Sanitary Sewer Systems.
(http://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2013/wqo2013_0058exec.pdf).
- e. The Discharger shall provide a report to the Colorado River Basin Water Board upon determining that the treatment plant's monthly average flow rate for any month exceeds 80 percent of the design treatment or disposal capacity. The report should indicate what steps, if any the Discharger intends to take to provide for the expected wastewater treatment capacity necessary when the plant reaches design capacity.
- f. Prior to any change in ownership or management of this operation, the Discharger shall transmit a copy of this Order to the succeeding owner/operator, and forward a copy of the transmittal letter to the Colorado River Basin Water Board. The new owner/operator shall apply for this Permit.
- g. Prior to any modifications in this facility, which would result in material change in the quality or, quantity of wastewater treated or discharged, or any material change in the location of discharge, the Discharger shall report all pertinent information in writing to the Colorado River Basin Water Board and if required by the Colorado River Basin Water Board obtain revised requirements before any modifications are implemented.
- h. Adequate measures shall be taken to assure that flood or surface drainage waters do not erode or otherwise render portions of the discharge facilities inoperable.

- i. This Order does not authorize violation of any federal, state, or local laws or regulations.
- j. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this facility, may subject the Discharger to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Discharger to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
- k. In the event the Discharger does not comply or will be unable to comply for any reason, with any prohibition, maximum daily, average weekly, average monthly, instantaneous maximum or instantaneous minimum, or receiving water limitation of this Order, the Discharger shall notify the Colorado River Basin Water Board by email to RB7-coloradoriver@waterboards.ca.gov within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing within five days, unless the Colorado River Basin Water Board waives confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and, prevent recurrence including, where applicable, a schedule of implementation. Other noncompliance requires written notification as above at the time of the normal monitoring report.

B. Monitoring and Reporting Program (MRP) Requirements

The Discharger shall comply with the MRP and future revisions thereto, in Attachment E of this Order. This MRP may be modified by the Executive Officer at any time during the term of this Order, and may include an increase in the number of parameters to be monitored, the frequency of the monitoring or the number and size of samples to be collected or minor clarifications on MRP requirements. Any increase in the number of parameters to be monitored, the frequency of the monitoring or the number and size of samples to be collected may be reduced back to the levels specified in the original MRP at the discretion of the Executive Officer.

C. Special Provisions

1. Reopener Provisions

- a. This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity, monitoring requirements on internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data.
- b. This Order may be modified, rescinded and reissued, for cause. The filing of a request by the Discharger for an Order modification, rescission and reissuance, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. Causes for modification include the promulgation of new regulations, modification of land application plans, or modification in sludge use or disposal practices, or adoption of new regulations by the State Water Board or the Colorado River Basin Water Board, including revisions to the Basin Plan.

- c. The CWA requires the Colorado River Basin Water Board to modify, or terminate and reissue, the NPDES permit if a discharger must implement a pretreatment program. Public notice and a comment period are mandatory for these actions.
- d. This Order may be reopened and the Whole Effluent Toxicity (WET) Requirements, contained in section V of the MRP, may be modified to address changes to U.S. EPA or State Water Board policies or guidance regarding the testing or reporting requirements for WET testing.
- e. In accordance with 40 C.F.R. parts 122 and 124, this permit may be modified to include effluent limitations or permit conditions to address chronic toxicity in the effluent or receiving waterbody, as a result of the discharge; or to implement new, revised, or newly interpreted water quality standards applicable to chronic toxicity.
- f. If a statewide policy for total residual chlorine is adopted during the term of this Order, this Order may be reopened to include a revised reporting level to determine compliance with effluent limitations for total residual chlorine for discharges consistent with the statewide policy.
- g. If a statewide policy for nutrients is adopted during the term of this Order, this Order may be reopened to include a revised reporting level to determine compliance with effluent limitations for nutrients for discharges consistent with the statewide policy.
- h. TMDLs for CWA Section 303(d) listed impairments of Imperial Valley Drains and their tributary waters are to be developed by the Colorado River Basin Water Board. The permit may be reopened and modified to include appropriate requirements necessary to fully implement the approved TMDLs, if needed.

2. Special Studies, Technical Reports and Additional Monitoring Requirements

a. Whole Effluent Toxicity Requirements

For compliance with the Basin Plan's narrative toxicity objective, this Order requires the Discharger to conduct whole effluent toxicity (WET) testing, as specified in MRP section V. Furthermore, this Provision requires the Discharger to investigate the causes of, and identify corrective actions to reduce or eliminate effluent toxicity. If the discharge exceeds the numeric toxicity monitoring triggers specified in section V.C of the MRP, this Order requires the Discharger to initiate accelerated WET testing. If the Discharge exceeds the numeric toxicity monitoring triggers during the accelerated WET testing, the Discharger is required to initiate a Toxicity Reduction Evaluation (TRE) and Toxicity Identification Evaluation (TIE) in accordance with an approved TRE Work Plan. A TRE is a site-specific study conducted in a stepwise process to identify the source(s) of toxicity, evaluate effective control measures for effluent toxicity, and confirm the reduction in effluent toxicity. This Provision includes requirements for the Discharger to develop and submit a TRE Work Plan and, if necessary, initiate accelerated WET testing and a TRE/TIE.

b. Chronic Whole Effluent Toxicity.

The Discharger developed and submitted to the Colorado River Basin Water Board a TRE Work Plan to comply with the requirements of Order R7-2010-0020, as amended by Special Order R7-2011-0018. The Discharger shall review and update the existing TRE Work Plan on an annual basis. The Discharger shall submit the updated TRE Work Plan with each Annual Report. The TRE Work Plan shall outline the procedures for identifying the source(s) of, and reducing or eliminating effluent toxicity. The TRE Work Plan must be developed in accordance with U.S. EPA guidance provided in manuals EPA/600/2-88/070 (industrial) or EPA/833B-99/002

(municipal) and be of adequate detail to allow the Discharger to immediately initiate the TRE Work Plan upon notification from the WET testing laboratory of effluent toxicity. This plan shall describe the steps the Discharger intends to follow in the event that toxicity is detected, and should include at a minimum:

- i. A description of the investigation and evaluation techniques that would be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency.
- ii. A description of methods for maximizing in-house treatment system efficiency, good housekeeping practices, and a list of all chemicals used in operations at the facility.
- iii. If a Toxicity Identification Evaluation (TIE) is necessary, an indication of who would conduct the TIEs (i.e., an in-house expert or outside contractor).
- iv. Specific actions the Discharger will take to mitigate the impact of the discharge and prevent the recurrence of toxicity; and
- v. A schedule for these actions.

c. Translator Study

Should the Discharger wish to use a translator for metals and selenium other than the default U.S. EPA conversion factors listed in Tables 2 and 3 of the California Toxic Rule (CTR), the Discharger shall submit a written request to the Executive Officer. Otherwise, the U.S. EPA conversion factors shall remain the default standard used when calculating water quality-based effluent limitations for selenium and metals. U.S. EPA has developed a guidance manual, EPA 823-B-96-007, June 1996, entitled, "The Metals Translator: Guidance for Calculating a Total Recoverable Permit Limit from a Dissolved Criterion."

d. Discharge Monitoring Report-Quality Assurance (DMR-QA) Study

Under the authority of section 308 of the CWA (33 U.S.C. § 1318), U.S. EPA requires major Permittees under the NPDES Program to participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by U.S. EPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board's Quality Assurance Program Officer. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to U.S. EPA's DMR-QA Coordinator and Quality Assurance Manager.

e. **Total Dissolved Solids Study**

The Colorado River Basin Water Board required the Discharger to conduct a study and evaluate what programs and actions the Discharger is implementing to reduce TDS discharges into the receiving water body.

The Discharger is aware that excessive TDS loading and infiltration of high TDS groundwater have negative consequences, including environmental challenges, increased treatment costs, and detriment for any future water reuse. Based upon budget availability, the City is going to continue its wastewater facility rehabilitation and replacement program to minimize infiltration into the sewer collection system.

No further action on the Total Dissolved Solids Study is required.

3. Best Management Practices and Pollution Prevention

a. **Pollutant Minimization Program**

The Discharger shall develop and conduct a Pollutant Minimization Program (PMP) as further described below when there is evidence (e.g., sample results reported as "Detected, but Not Quantified" (DNQ) when the effluent limitation is less than the Method Detection Limit (MDL), sample results from analytical methods more sensitive than those methods required by this Order, presence of whole effluent toxicity, health advisories for fish consumption, or results of benthic or aquatic organism tissue sampling) that a priority pollutant is present in the effluent above an effluent limitation and either:

- i. A sample result is reported as DNQ and the effluent limitation is less than the Reporting Level (RL); or
- ii. A sample result is reported as Not Detected (ND) and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in MRP section X.B.5.

The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Colorado River Basin Water Board:

- i. An annual review and semi-annual monitoring of potential sources of the reportable priority pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling;
- ii. Quarterly monitoring for the reportable priority pollutant(s) in the influent to the wastewater treatment system;
- iii. Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable priority pollutant(s) in the effluent at or below the effluent limitation;
- iv. Implementation of appropriate cost-effective control measures for the reportable priority pollutant(s), consistent with the control strategy; and
- v. An annual status report that shall be sent to the Colorado River Basin Water Board including:
 - (a) All PMP monitoring results for the previous year;
 - (b) A list of potential sources of the reportable priority pollutant(s);
 - (c) A summary of all actions undertaken pursuant to the control strategy; and
 - (d) A description of actions to be taken in the following year.

b. Storm Water

Enrollment under Water Quality Order 2014-0057-DWQ, NPDES General Permit CAS000001, for Discharges of Storm Water Associated with Industrial Activities is required for facilities used in the storage, treatment, recycling, and reclamation of municipal or domestic sewage, including land dedicated to the disposal of sewage sludge that are located within the confines of the facility with a design flow of 1 MGD or more or are required to have an approved pretreatment program under 40 C.F.R. part 403.

4. Construction, Operation and Maintenance Specifications

a. Treatment Basins

- i. A minimum depth of freeboard of two (2) feet shall be maintained at all times in earthen basins which do not have a protective liner or structural design. For basins with liner or structural design a 1-foot freeboard is acceptable.
- ii. The treatment basins shall be managed to control breeding of mosquitoes, in particular:
 - (a) An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface;
 - (b) herbicides; and
 - (c) Dead algae, vegetation, and debris shall not accumulate on the water surface.
- iii. The treatment basins shall be maintained so they will be kept in aerobic conditions.
- iv. On-site wastes shall be strictly confined to the lands specifically designated for the disposal operation.
- v. Public contact with un-disinfected wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.
- vi. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal area.

b. Facility and Treatment Operation

- i. The Discharger shall, at all times, properly operate and maintain all systems and components of collection, treatment and control which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance includes effective performance, adequate process controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems when necessary to achieve compliance with the conditions of this Order. All systems, both in-service and reserved, shall be inspected and maintained on a regular basis. Records shall be kept of the inspection results and maintenance performed and made available to the Colorado River Basin Water Board upon demand.
- ii. Temporary power or adequate storage capacity shall be provided to maintain the plant in operation in the event of commercial power failure.
- iii. Adequate measures shall be taken to assure that unauthorized persons are effectively excluded from contact with the wastewater disposal facilities.

- iv. The Discharger shall implement acceptable operation and maintenance at the facility so that needed repair and maintenance are performed in a timely manner.

c. Anti-degradation Analysis and Engineering Report for Significant Expansion

Discharges from a new or an existing facility that will undergo significant expansion¹ within the next 5 years shall be required to submit an anti-degradation analysis report to the Colorado River Basin Water Board's Executive Officer for review and approval. The anti-degradation analysis report shall be developed in accordance with the State Anti-degradation Policy (Resolution No. 68-16) and the Federal Anti-degradation Policy (40 C.F.R. section 131.12). The report shall consider any potential impacts the discharge may have on the receiving water quality and the receiving water body's designated beneficial uses, as defined in the Colorado River Basin Water Board's Basin Plan. In addition, the report shall include information on the quality of the proposed discharge and evaluate the potential impacts of the discharge; provide CEQA documentation for the proposed project; a summary that identifies whether the proposed discharge will result in degradation of water quality; and a certification that satisfies both the Federal and State anti-degradation policies.

d. Spill Response Plan

The Discharger developed and submitted to the Colorado River Basin Water Board a Spill Response Plan (SRP) to comply with the requirements of Order R7-2010-0020, as amended by Special Order R7-2011-0018. The Discharger shall review and update the existing SRP on an annual basis. The Discharger shall submit the updated SRP with each Annual Report. The Discharger shall make the SRP available for staff review during Colorado River Basin Water Board inspections. The Discharger shall ensure that all operating personnel are familiar with the contents of the SRP. A copy of the SRP shall be maintained at the site and shall be accessible to all operating personnel.

5. Special Provisions for Municipal Facilities (POTWs Only)

a. Sludge Requirements.

i. General Requirements

- (a) In the event that sludge is generated at the treatment facility, all sludge generated at the treatment facility shall be used or disposed of in compliance with the applicable portions of:
 - (1) 40 C.F.R. part 503: for sludge land applied, placed in surface disposal sites (dedicated land disposal sites, monofill, or sludge-only parcel at municipal landfill), or incineration.
 - (2) 40 C.F.R. part 258: for sludge disposed in municipal solid waste landfills (with other materials).
 - (3) 40 C.F.R. part 257: for all sludge use and disposal practices not covered under 40 C.F.R. parts 503 or 258.

¹ Significant expansion shall be considered an increase in permitted design flow of greater than 10% or changes to the Facility and/or changes in the nature and character of the discharge that may result in an incremental increase in pollutants discharged to the receiving water body of greater than 10% of the permitted discharge rates. When a new or existing facility undergoes significant expansion, the discharge shall be evaluated on a pollutant by pollutant basis.

- (b) In the event that sludge is generated at the treatment facility, the Discharger is responsible for assuring that all sludge generated at the treatment facility is used or disposed of in accordance with these rules, whether the Discharger uses or disposes of the sludge itself or transfers the sludge to another party for further treatment and use, or disposal. The Discharger is responsible for informing subsequent preparers, appliers, and disposers of the requirements that they must meet under these rules.
- (c) In the event that sludge is generated at the treatment facility, the Discharger shall assure that haulers transporting sewage sludge from the treatment facility for treatment, storage, use, or disposal take appropriate measures to keep the sludge contained. Trucks hauling sewage sludge that is not Class A as defined in 40 C.F.R. section 503.32(a) shall be covered.
- (d) In the event that sludge is generated at the treatment facility, any sludge treatment, disposal, or storage site shall have adequate procedures to restrict public access and access by domestic animals.
- (e) In the event that sludge is generated at the treatment facility, sludge treatment, disposal or storage sites shall have facilities adequate to divert surface runoff from adjacent areas, to protect the site boundaries from erosion, and to prevent any conditions that would cause drainage from the sludge to escape from the site. Adequate protection is defined as protection from at least a 100-year storm.
- (f) In the event that sludge is generated at the treatment facility, for sewage sludge that is applied to land in accordance with 40 C.F.R part 503 Subpart B, the wastewater treatment facility shall have adequate screening at the plant headworks and/or at the sludge treatment units to ensure that all pieces of metals, plastics, glass, and other inert objects with a diameter of greater than 3/8" are removed.
- (g) Collected screenings, sludge, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with State Water Board and California Department of Resources Recycling and Recovery's (CalRecycle) consolidated regulations in Title 27 of the California Code of Regulations (CCR).

ii. **Notification and Reporting Requirements**

- (a) The Discharger shall provide a plan as to the method, treatment, handling, and disposal of sludge that is consistent with all State and federal laws and regulations and obtain prior written approval from the Colorado River Basin Water Board's Executive Officer specifying location and method of disposal, before disposing of treated or untreated sludge, or similar solid waste materials using an alternative method than that described in the Findings of the Order. In addition, the Discharger's plan for managing sludge is described in section II.A of the associated Fact Sheet.
- (b) If sludge generated at the treatment facility is stored for over two years from the time it is generated, the Discharger shall ensure compliance with all the requirements for surface disposal under 40 C.F.R. part 503 Subpart C, or shall submit written notification in accordance with 40 C.F.R. section 503.20(b) to the U.S. EPA and the Colorado River Basin Water Board demonstrating the need for longer temporary storage.

- (c) The Discharger shall maintain a permanent log of all solids hauled away from the treatment facility for use/disposal elsewhere and shall provide a summary of the volume, type (screenings, grit, raw sludge, digested sludge), use (agricultural, composting, etc.), and the destination in accordance with the MRP of this Order. The sludge that is stockpiled at the treatment facility shall be sampled and analyzed for those constituents listed in the sludge monitoring section of the MRP of this Order and as required by 40 C.F.R. part 503. The results of the analyses should be submitted to the Colorado River Basin Water Board as part of the MRP.

b. Pretreatment

- i. In the event that (i) the facility has a treatment capacity greater than 5 MGD and Industrial Users [40 C.F.R. § 403.3(j)] are discharging pollutants which Pass Through [40 C.F.R. § 403.3(p)] or Interfere [40 C.F.R. § 403.3(k)] with the operation of the wastewater treatment facility or are otherwise subject to National Pretreatment Standards [40 C.F.R. § 403.3(l)], (ii), Section 2233 of title 23 of the CCRs requires the facility to have and enforce an adequate pretreatment program, or (iii) the Colorado River Basin Water Board or its Executive Officer determines that other circumstances warrant in order to prevent Interference with the wastewater treatment facility or Pass Through, then:
 - (a) The Discharger shall be responsible for the compliance with all pretreatment requirements contained in 40 C.F.R part 403, and shall be subject to enforcement actions, penalties, and other remedies by the U.S. EPA, or the Colorado River Basin Water Board, as provided in the CWA.
 - (b) Within one year of notification that a pretreatment program is required, the Discharger shall submit a formal Pretreatment Program Submission for approval by the Colorado River Basin Water Board.

The Discharger must seek approval of its Pretreatment Program from the Colorado River Basin Water Board subject to Provision VI.C.1.c of this Order in the event a Pretreatment Program is developed.

c. Collection Systems

- i. The Discharger's collection system is part of the system that is subject to this Order. As such, the Discharger must properly operate and maintain its collection system (40 C.F.R. § 122.41(e)). The Discharger must report any non-compliance (40 C.F.R. § 122.41(l)(6) and (7)) and mitigate any discharge from the collection system in violation of this Order (40 C.F.R. § 122.41(d)). See the Order at Standard Provision VI.A.2.e and Attachment D, subsections I.C, I.D, V.E, and V.H.
- ii. To provide a consistent, statewide regulatory approach to address Sanitary Sewer Overflows (SSOs), the State Water Resources Control Board (State Water Board) adopted Statewide General Waste Discharge Requirements (WDRs) for Sanitary Sewer Systems, Water Quality Order No. 2006-0003 (Sanitary Sewer Order) on May 2, 2006. The Sanitary Sewer Order requires public agencies that own or operate sanitary sewer systems to develop and implement sewer system management plans and report all SSOs to the State Water Board's online SSO database. The Discharger is enrolled under the SSO Order. The WDID number is 7SSO10519.

6. Other Special Provisions

- a. The Discharger shall submit the Electronic Self-Monitoring Reports (eSMRs) for reporting periods September and Third Quarter 2015 in the SMR module for Order No. R7-2010-0020, as amended by Special Order R7-2011-0018, in the California Integrated Water Quality System (CIWQS) Program.
- b. The Discharger may be required to submit technical reports as directed by the Colorado River Basin Water Board's Executive Officer.
- c. The Discharger shall exclude from the wastewater treatment plant any liquid or solid waste that could adversely affect the plant operation or effluent quality. The excluded liquid or solid waste shall be disposed of in accordance with applicable regulations.

7. Compliance Schedules

- a. **Deliverables and Due Dates.** The Discharger shall comply with the following compliance schedule as summarized in Table 6:

Table 6. Compliance Schedule

Activity	Description	Due Date
Spill Response Plan VI.C.4.c.	The Discharger shall update the Spill Response Plan (SRP). The SRP shall be updated annually and available for inspection.	Annual updates submitted yearly to Colorado River Basin Water Board with Annual Report.
TRE Workplan VI.C.2.b.	The Discharger shall update the TRE Work Plan. The Work Plan should include a description of steps the Discharger will take in the event toxicity is detected. The Work Plan should describe investigation and evaluation techniques used to identify sources of toxicity; method for maximizing in-house efficiency; and identify the party who will conduct the TIE.	Annual updates submitted yearly to Colorado River Basin Water Board with Annual Report.
DMR-QA Study VI.C.2.d	The Discharger shall conduct a DMR-QA Study to evaluate the analytical ability of laboratories that routinely perform or support self-monitoring analyses. The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by U.S. EPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board.	Annual updates submitted yearly to the State Water Board's Quality Assurance Officer.

Activity	Description	Due Date
Sludge Disposal Notification and Plan VI.C.5.a.ii.(a)	The Discharger shall provide a plan as to the method, treatment, handling, and disposal of sludge that is consistent with all State and federal laws and regulations and obtain prior written approval from the Colorado River Basin Water Board specifying location and method of disposal, before disposing of treated or untreated sludge, or similar solid waste materials using an alternative method than that described in the Findings of the Order.	Prior to disposal of sludge using alternative method.
Anti-degradation Analysis and Engineering Report for Significant Expansion VI.C.4.b	Discharges from a new or an existing facility that will undergo significant expansion within the next 5 years shall be required to submit an anti-degradation analysis report to the Colorado River Basin Water Board's Executive Officer for review and approval. The anti-degradation analysis report shall be developed in accordance with the State Anti-degradation Policy (Resolution No. 68-16) and the Federal Anti-degradation Policy (40 C.F.R. section 131.12). The report shall consider any potential impacts the discharge may have on the receiving water quality and the receiving water bodies designated beneficial uses, as defined in the Colorado River Basin Water Board's Basin Plan.	Prior to start of construction of significant changes to the treatment plant and to be submitted to Colorado River Basin Water Board's Executive Officer.
Pollutant Minimization Program (PMP) VI.C.3.a	The Discharger shall develop a PMP when there is evidence a priority pollutant is present in the effluent above an effluent limitation and either the sample result is reported as DNQ and the effluent limitation is less than the RL; or a sample result is reported as ND and the effluent limitation is less than the MDL. The PMP shall include an annual review of monitoring pollutant sources, quarterly influent monitoring for the priority pollutants, submittal of a pollutant control strategy, and an annual status report which will include PMP monitoring results for the previous year, a list of pollutant sources, summary of actions taken pursuant to the control strategy, and a description of actions to be taken in the following year.	Upon evidence of priority pollutant in effluent and to be submitted to Colorado River Basin Water Board.

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in section IV of this Order will be determined as specified below:

A. General.

Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined in the MRP and VII. of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the Reporting Level (RL).

B. Multiple Sample Data.

When determining compliance with an Average Monthly Effluent Limitation (AMEL), Average Weekly Effluent Limitation (AWEL) and Maximum Daily Effluent Limitation (MDEL) for pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

1. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
2. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

C. Average Monthly Effluent Limitation (AMEL).

If the average (or when applicable, the median determined by subsection B above for multiple sample data) of daily discharges over a calendar month exceeds the AMEL for a given parameter, this will represent a single violation, though the Discharger will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month), where no data is available to show compliance. If only a single sample is taken during the calendar month and the analytical result for that sample exceeds the AMEL, the Discharger will be considered out of compliance for that calendar month. The Discharger will only be considered out of compliance for days when the discharge occurs. For any one calendar month during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar month.

D. Average Weekly Effluent Limitation (AWEL).

If the average (or when applicable, the median determined by subsection B above for multiple sample data) of daily discharges over a calendar week exceeds the AWEL for a given parameter, this will represent a single violation, though the Discharger will be considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance where no data is available to show compliance. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Discharger will be considered out of compliance for that calendar week. The Discharger will only be considered out of compliance for days when the discharge occurs. For any one calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week. The analytical result for comparison of the AWEL shall be reported on the ending Saturday of the calendar weeks within the reporting month.

For the beginning of the month, if samples are collected during the calendar week in the previous month and the ending Saturday is in the Calendar Month of reporting, the Discharger shall report the results of the samples collected in the previous week and report the analytical result on the ending Saturday of the Calendar Month of reporting.

For the end of the month, if the ending Saturday falls on the following month no analytical result for comparison of the AWEL is required, however, any samples collected during the last calendar week shall be reported.

E. Maximum Daily Effluent Limitation (MDEL).

If a daily discharge (or when applicable, the median determined by subsection B above for multiple sample data of a daily discharge) exceeds the MDEL for a given parameter, the Discharger will be considered out of compliance for that parameter for that 1 day only within the reporting period. For any 1 day during which no sample is taken, no compliance determination can be made for that day.

For multiple samples collected in a calendar day, the maximum daily value is the average of the samples collected in a calendar day, or when applicable, the median as determined by subsection B, above.

F. Instantaneous Minimum Effluent Limitation.

If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in two instances of non-compliance with the instantaneous minimum effluent limitation). There are no mass limits for instantaneous minimum effluent limitations.

G. Instantaneous Maximum Effluent Limitation.

If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, the Discharger will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in two instances of non-compliance with the instantaneous maximum effluent limitation). There are no mass limits for instantaneous maximum effluent limitations.

H. Effect of Conducting a Pollutant Minimization Program (PMP).

If a sample result for a priority pollutant, or the arithmetic mean or median of multiple sample results is below the RL, and there is evidence that the priority pollutant is present in the effluent above an effluent limitation and the Discharger conducts a PMP for the priority pollutant (as described in Provision VI.C.3.a.), the Discharger shall not be deemed out of compliance.

I. Water Quality Based Effluent Limitations.

1. In accordance with section 2.4.5 of the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays and Estuaries of California (State Implementation Policy or SIP), compliance with water quality-based effluent limitations shall be determined as follows:
 - a. Dischargers shall be deemed out of compliance with an effluent limitation if the concentration of a priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reported Minimum Level (ML).
 - b. When determining compliance with an average monthly effluent limitation and more than one sample result is available in a month, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of DNQ or ND. In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:

- i. The data set shall be ranked from low to high, reported ND determinations lowest, DNQ determinations next, and followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
- ii. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than a DNQ.
- iii. If a sample result, or the arithmetic mean or median of multiple sample results, is below the reported ML, and there is evidence that the priority pollutant is present in the effluent above an effluent limitation and the Discharger conducts a PMP, the Discharger shall not be deemed out of compliance.

J. Mass and Concentration Limitation

Compliance with mass and concentration effluent limitations for the same parameter shall be determined separately with their respective limitations. When the concentration of a constituent in an effluent sample is determined to be ND or DNQ, the corresponding mass emission rate (MER) determined from that sample concentration shall also be reported as ND or DNQ.

K. Percent Removal

Compliance with the secondary treatment standard for monthly average percent removal of biochemical oxygen demand and total suspended solids, pursuant to 40 C.F.R. part 133, shall be determined separately for each wastewater treatment facility discharging through an outfall. For each wastewater treatment facility, the monthly average percent removal is the average of the calculated daily discharge percent removals only for days on which the constituent concentrations is monitored in both the influent and effluent of the wastewater treatment facility at locations specified in the MRP (Attachment E) within a calendar month.

The percent removal for each day (Daily Percent Removal) shall be calculated according to the following equation:

$$\text{Daily Percent Removal} = \frac{(\text{Daily Influent Concentration} - \text{Daily Effluent Concentration})}{\text{Daily Influent Concentration}} * 100$$

The percent removal for the month (Monthly Percent Removal) shall be calculated according to the following equation:

$$\text{Monthly Percent Removal} = \frac{(\text{Sum of the Daily Percent Removal})}{\text{Number of Daily Percent Removal Values}}$$

L. Chronic Toxicity Narrative Language.

Compliance with narrative effluent limitations established in the Order shall be determined as follows:

Reasonable potential for toxicity for this discharge has not been determined, hence effluent chronic toxicity limit based on WET tests do not exist. However, compliance with narrative effluent limitations established in the Order comprises of chronic toxicity triggers. The chronic toxicity permit triggers for this discharge are:

1. Any chronic toxicity test result that exceeds 2 chronic toxicity units (TUc) or a three (3)-sample median (consecutive samples) that exceeds 1 TUc shall trigger accelerated WET testing; or
2. Any single concentration toxicity test where statistical significant difference exists between the control and in-stream waste concentration (IWC) is considered a Fail result. For this discharge, the IWC is 100 percent effluent. A Pass result indicates no toxicity at the IWC, and a Fail result indicates toxicity at the IWC. The Discharger must report either a Pass or a Fail and the percent effect as required in the Monitoring and Reporting Program, section V. If a result is reported as a Fail, the Discharger must follow the requirements in MRP, section V.D., Accelerated Toxicity Testing and TRE/TIE Process. Failure to initiate an accelerated monitoring schedule or conduct a TRE/TIE may result in appropriate enforcement action.

M. Bacterial Effluent Limitations.

Compliance with the bacterial effluent limitations established in section IV.A.1.d of this Order shall be determined as follows:

1. If the calculated geometric mean bacterial concentrations for *E. coli* or fecal coliform exceed the 30-day geometric mean effluent limitations summarized in the Effluent Limitations and Discharge Requirements section IV.A.1.d of this Order or if more than ten percent of total samples for fecal coliform exceed 400 MPN per 100 milliliters, this will represent a single violation of the water quality-based effluent limitation for bacteria and the Discharger will be considered out of compliance for the month in which the samples were collected.
 - (a) In cases where the calculated geometric mean bacterial concentrations for *E. coli* and fecal coliform exceed the 30-day geometric mean effluent limitations, *E. coli* shall be noted as the violation.
 - (b) If the calculated geometric mean bacterial concentration for only one bacterial indicator exceeds the effluent limitation, that calculated geometric mean bacterial concentration for indicator parameter shall be noted as the violation (i.e., *E. coli* or fecal coliform).
2. If the bacterial concentrations for *E. coli* or fecal coliform exceed the maximum bacterial densities summarized in the Effluent Limitations and Discharge Requirements section IV.A.1.d of this Order, this will represent a single violation of the water quality-based effluent limitation for bacteria and the Discharger will be considered out of compliance for the day in which the sample(s) were collected.
 - (a) In cases where individual bacterial concentrations *E. coli*, and fecal coliform exceed the maximum bacterial densities in the effluent limitations, *E. coli* shall be noted as the violation.
 - (b) If only the bacterial concentration for only one bacterial indicator exceeds the effluent limitations, that bacterial indicator parameter shall be noted as the violation (i.e., *E. coli* or fecal coliform).

N. Significant Figures

The Discharger shall report monitoring and calculation results with regard to significant figures consistent with tabulated values in Table 5 (Effluent Limitations).

ATTACHMENT A – DEFINITIONS

Acute Toxicity Test

Acute toxicity test is a test to determine the concentration of effluent or ambient waters that causes an adverse effect (usually mortality) on a group of test organisms during a short-term exposure (e.g., 24, 48, or 96 hours). Acute toxicity is determined using statistical procedures (e.g., point estimates or a t-test).

Ambient Toxicity

Ambient toxicity is measured by a toxicity test on a sample collected from a receiving waterbody.

Annual Average Effluent Limitation

The highest allowable average of monthly discharges over a calendar year, calculated as the sum of all monthly discharges measured during a calendar year divided by the number of monthly discharges measured during that year.

Arithmetic Mean (μ)

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

$$\text{Arithmetic mean} = \mu = \Sigma x / n \quad \text{where: } \Sigma x \text{ is the sum of the measured ambient water concentrations, and } n \text{ is the number of samples.}$$

Average Monthly Effluent Limitation (AMEL)

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

For the AMEL concentration limit, it is the sum of the measured sample values divided by the number of samples for the month.

For the AMEL mass loading limit, it is the sum of the product of the flow rate (mgd) x measured sample value (mg/L) x 8.34 divided by the number of samples for the month.

Average Weekly Effluent Limitation (AWEL)

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

Best Management Practices (BMPs)

BMPs are methods, measures, or practices designed and selected to reduce or eliminate the discharge of pollutants to surface waters from point and non-point discharges including storm water. BMPs include structural and non-structural controls, and operation and maintenance procedures, which can be applied before, during, and/or after pollution producing activities.

Bioaccumulative

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

Biosolids

Biosolids refer to non-hazardous sewage sludge as defined in 40 C.F.R. section 503.9.

Carcinogenic

Pollutants are substances that are known to cause cancer in living organisms.

Chronic Toxicity Tests

Chronic toxicity tests measure the sub-lethal effects of a discharge (e.g. reduced growth or reproduction). Certain chronic toxicity tests include an additional measurement of lethality.

Coefficient of Variation (CV)

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

Criteria Continuous Concentration (CCC)

Criteria Continuous Concentration equals the highest concentration of a pollutant to which aquatic life can be exposed for an extended period of time (e.g., 4 days) without deleterious effects.

Criteria Maximum Concentration (CMC)

Criteria Maximum Concentration equals the highest concentration of a pollutant to which aquatic life can be exposed for a short period of time (e.g., 1 hour) without deleterious effects.

Daily Discharge

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

Detected, but Not Quantified (DNQ)

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

Dilution Credit

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

Domestic Sewage

Domestic Sewage is waste and wastewater from humans or household operations that is discharged to or otherwise enters a treatment works.

Effect Concentration (EC)

Effect concentration is a point estimate of the toxicant concentration that would cause an observable adverse effect (e.g., mortality, fertilization). EC25 is a point estimate of the toxicant concentration that would cause observable 25% adverse effect as compared to the control test organisms.

Effluent Concentration Allowance (ECA)

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in U.S. EPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

Enclosed Bays

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

Estimated Chemical Concentration

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

Estuaries

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

Existing Discharger

Any Discharger that is not a new Discharger. An existing Discharger includes an "increasing Discharger" (i.e., an existing Facility with treatment systems in place from its current discharge that is or will be expanding, upgrading, or modifying its existing permitted discharge after the effective date of the State Implementation Policy).

Geometric Mean

Geometric mean, is a measure of the central tendency of a data set that minimizes the effects of extreme values. The geometric mean used for determining compliance with bacterial standards is calculated with the following equation:

Geometric Mean = $(C_1 \times C_2 \times \dots \times C_n)^{1/n}$ where n = the number of days samples were collected during the period, and C = the concentration of bacteria (CFU*/100 mL) found on each day of sampling.

*Effluent limitations for bacterial density are expressed in units of a Most Probable Number per 100 milliliters (MPN/100 ml). This calculation of geometric mean is also applicable and shall be used to determine compliance with bacterial effluent limitations.

Group I Pollutants

The list of pollutants is based on Appendix A to 40 C.F.R § 123.45. The State Water Resources Control Board enforcement policy located at

http://www.waterboards.ca.gov/water_issues/programs/enforcement/docs/enf_policy_final111709.pdf provides the list in Appendix C: Group 1 Pollutants.

Group 2 Pollutants

The list of pollutants is based on Appendix A to 40 C.F.R § 123.45. The State Water Resources Control Board enforcement policy located at

http://www.waterboards.ca.gov/water_issues/programs/enforcement/docs/enf_policy_final111709.pdf provides the list in Appendix D: Group 2 Pollutants.

Hypothesis Testing

Hypothesis testing is a statistical approach (e.g., Dunnett's procedure) for determining whether a test concentration is statistically different from the control. Endpoints determined from hypothesis testing are no observed effect concentration (NOEC) and lowest observed effect concentration (LOEC).

Incompletely Mixed Discharge

A discharge that contributes to a condition that does not meet the meaning of a completely-mixed discharge condition.

Infeasible

Infeasible means not capable of being accomplished in a successful manner within a reasonable period of time, taking into account economic, environmental, legal, social, and technological factors.

Inhibition Concentration

Inhibition concentration is a point estimate of the toxicant concentration that would cause a given, percent reduction in a non-lethal biological measurement (e.g., reproduction or growth), calculated from a continuous model (i.e., Interpolation Method). For example, IC25 is a point estimate of the toxicant concentration that would cause a 25 percent reduction in a non-lethal biological measurement.

Inland Surface Waters

All surface waters of the state that do not include the ocean, enclosed bays, or estuaries.

Instantaneous Maximum Effluent Limitation

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

Instantaneous Minimum Effluent Limitation

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

In-Stream Waste Concentration

In-stream waste concentration (IWC) is the concentration of a toxicant or effluent in the receiving water after mixing (the inverse of the dilution factor). A discharge of 100% effluent is considered the IWC for this discharge.

LC50

LC50 (lethal concentration, 50%) is the toxicant or effluent concentration that would cause death to 50 percent of the test organisms.

Load Allocation

The portion of a receiving water's total maximum daily load that is allocated to one of its non-point sources of pollution or to natural background sources.

Lowest Observed Effect Concentration

Lowest observed effect concentration (LOEC) is the lowest concentration of an effluent or toxicant that results in statistically significant adverse effects on the test organisms (i.e., where the values for the observed endpoints are statistically different from the control).

Maximum Daily Effluent Limitation (MDEL)

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

Median

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median = $X_{(n+1)/2}$. If n is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the $n/2$ and $n/2+1$).

Method Detection Limit (MDL)

MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 C.F.R. part 136, Attachment B, revised as of July 3, 1999.

Minimum Level (ML)

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

Minimum Significant Difference (MSD)

Minimum significant difference is the magnitude of difference from control where the null hypothesis is rejected in a statistical test comparing a treatment with a control. MSD is based on the number of replicates, control performance, and power of the test.

Mixing Zone

The Colorado River Basin Water Board does not have a mixing zone policy in the Basin Plan. Therefore, in order for a mixing zone to be allowed in the Colorado River Basin Region, it would be only pursuant to a State policy. The State Implementation Policy (SIP) allows a mixing zone for priority pollutants and toxicity. Accordingly, a mixing zone applies to the Colorado River Basin Region under this State policy.

The SIP requires a mixing zone analysis be completed before any dilution credit is granted. Following completion of the mixing zone study, the Colorado River Basin Water Board must reconsider the receiving water limitations to ensure that they are as stringent as necessary to fully protect the receiving water.

Municipality

Municipality means a city, town, borough, county, parish, district, association, or other public body created by or under State law and having jurisdiction over disposal of sewage, industrial wastes, or other wastes, or an Indian tribe or an authorized Indian tribal organization, or a designated and approved management agency under section 208 of CWA.

New Discharger

New Discharger includes any building, structure, Facility, or installation from which there is, or may be, a discharge of pollutants, the construction of which commenced after the effective date of the State Implementation Policy.

No Observed Effect Concentration (NOEC)

No observed effect concentration is the highest tested concentration of an effluent or toxicant that causes no observable adverse effect on the test organisms (i.e., the highest concentration of toxicant at which the values for the observed responses are not statistically different from the control).

Not Detected (ND)

Sample results which are less than the laboratory's MDL.

Objectionable Bottom Deposits

Objectionable Bottom Deposits are an accumulation of materials or substances on or near the bottom of a water body, which creates conditions that adversely impact aquatic life, human health, beneficial uses, or aesthetics. These conditions include, but are not limited to, the accumulation of pollutants in the sediments and other conditions that result in harm to benthic organisms, production of food chain organisms, or fish egg development. The presence of such deposits shall be determined by Colorado River Basin Water Board(s) on a case-by-case basis.

Ocean Waters

Not Applicable.

Percent Effect

The percent effect represents the difference between the response of the species at the IWC (i.e., 100% effluent) and the response in the control sample, relative to the control sample, as a percentage. The percent effect at IWC can be calculated as follows:

$$\text{Percent Effect} = \frac{(\text{Control Mean Response} - \text{IWC Mean Response})}{\text{Control Mean Response}} * 100$$

Persistent Pollutants

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

PET Tool

The PET tool is a Microsoft Excel file that allows you to configure your data into a format that CIWQS will understand and interpret correctly, which is the CIWQS Data Format, or CDR. You can open the PET Tool in Excel, configure it on the basis of your permit requirements, and then use the configured file as a template for entering data during the different reporting frequency and periods.

Pollutant Minimization Program (PMP)

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Colorado River Basin Water Board may consider cost effectiveness when establishing the requirements of a PMP. The

completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

Pollution Prevention

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board) or Colorado River Basin Water Board.

Potable Water

Water that is safe for drinking and cooking and is in compliance with the State Water Resources Control Board Division of Drinking Water or local county health department regulations.

Public Entity

Public Entity includes the Federal government or a state, county, city and county, city, district, public authority, or public agency.

Publicly Owned Treatment Works (POTW)

POTW means a treatment works as defined in 40 C.F.R. 212, which is owned by a State or municipality. This definition includes any devices and systems used in the storage, treatment, recycling and reclamation of municipal sewage or industrial wastes of a liquid nature. It also includes sewers, pipes, and other conveyances only if they convey wastewater to a POTW Treatment Plant. The term also means the municipality as defined in 40 C.F.R. 502(4), which has jurisdiction over the Indirect Discharges to and the discharges from such a treatment works.

Quality Assurance (QA)

Quality assurance is a practice in toxicity testing that addresses all activities affecting the quality of the final effluent toxicity data. QA includes practices such as effluent sampling and handling, source and condition of test organisms, equipment condition, test conditions, instrument calibration, replication, use of reference toxicants, recordkeeping, and data evaluation.

Quality Control (QC)

Quality control is the set of more focused, routine, day-to-day activities carried out as part of the overall QA program.

Reference Toxicant Test

Reference toxicant test is a check of the sensitivity of the test organisms and the suitability of the test methodology. Reference toxicant data are part of a routine QA/QC program to evaluate the performance of laboratory personnel and the robustness and sensitivity of the test organisms.

Replicate

Replicate is two or more independent organism exposures of the same treatment (i.e., effluent concentration) within a whole effluent toxicity test. Replicates are typically separate test chambers with organisms, each having the same effluent concentration.

Report of Waste Discharge

For the purposes of this Individual Board Order, references to the Report of Waste Discharge (ROWD) shall include the California Form 200, U.S. EPA forms and any other application information submitted to the Colorado River Basin Water Board.

Reporting Level (RL)

The RL is the ML (and its associated analytical method) chosen by the Discharger for reporting and compliance determination from the MLs included in this Order, including an additional factor if applicable as discussed herein. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Colorado River Basin Water Board either from Appendix 4 of the SIP in accordance with section 2.4.2 of the SIP or established in accordance with section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

Sample

Sample is a representative portion of a specific environmental matrix that is used in toxicity testing.

Satellite Collection System

The portion, if any, of a sanitary sewer system owned or operated by a different public agency than the agency that owns and operates the wastewater treatment facility that a sanitary sewer system is tributary to.

Serious Violation

For discharges of pollutants subject to the State Water Board's "Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California," or the "California Ocean Plan", where the effluent limitation for a pollutant is lower than the applicable Minimum Level, any discharge that: (1) equals or exceeds the Minimum Level; and (2) exceeds the effluent limitation by 40 percent or more for a Group 1 pollutant or by 20 percent or more for a Group 2 pollutant, is a serious violation for the purposes of California Water Code section 13385(h)(2).

For discharges of pollutants that are not subject to the State Water Board's "Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California," or the California Ocean Plan (e.g., pollutants that are not addressed by the applicable plan) where the effluent limitation for a pollutant is lower than the quantification limit specified or authorized in the applicable waste discharge requirements or monitoring requirements, any discharge that: (1) equals or exceeds the quantification limit; and (2) exceeds the effluent limitation by 40 percent or more for a Group 1 pollutant or by 20 percent or more for a Group 2 pollutant, is a serious violation for the purposes of California Water Code section 13385(h)(2).

Sewage Sludge

Sewage sludge is solid, semi-solid, or liquid residue generated during the treatment of domestic sewage in a treatment works. Sewage sludge includes, but is not limited to, domestic septage; scum or solids removed in primary, secondary, or advanced wastewater treatment processes; and a material derived from sewage sludge. Sewage sludge does not include ash generated during the firing of sewage sludge in a sewage sludge incinerator or grit and screenings generated during preliminary treatment of domestic sewage in a treatment works. Sewage sludge that has been classified as hazardous shall be disposed in accordance with 40 C.F.R. 261.

Sewage Sludge, Class A

Sewage Sludge to be classified Class A with respect to pathogens shall comply with the requirements in 40 C.F.R. 503.32(a)(2) and the requirements in either 40 C.F.R. 503.32(a)(3), (a)(4), (a)(5), (a)(6), (a)(7), or (a)(8).

Sewage Sludge, Class B

Sewage Sludge to be classified Class B with respect to pathogens shall comply with the requirements in either 40 C.F.R. 503.32(b)(2), (b)(3), or (b)(4).

Significant Difference

Significant difference is a statistically significant difference (e.g., 95 percent confidence level) in the means of two distributions of sampling results.

Significant Figures

Significant figures of a number are those digits that carry meaning contributing to its precision. When adding or subtracting values with different degrees of precision, the last digit retained is determined by the least precise number (i.e., the answer should contain no digits farther to the right of the least precise number). For example:

$$\begin{array}{r} 37.24 \\ +10.3 \\ \hline 47.54 \end{array} \text{ is rounded to } 47.5$$

When multiplying or dividing values with different degrees of precision, the number of significant figures in the answer equals that of the quantity that has the smallest number of significant figures. For example:

$$\overbrace{113.2}^4 \times \overbrace{1.43}^3 = \overbrace{161.876}^6 \text{ is rounded to } \overbrace{162}^3$$

Additional Information on significant figures.

- All nonzero digits are significant.
- Zeros between nonzero digits are significant (e.g., 1.005 mg has four significant figures).
- When a number ends in zeros to the right of a decimal point, they are significant (0.00500 has three significant figures).
- When a number ends in zeros that are not to the right of a decimal point, significant figures are indeterminable (e.g., 10300 kg).
- Only measurements have a limited number of significant figures. Given values, constants, etc. are assumed to have an infinite number of significant figures.

In addition, 40 C.F.R. part 136 specifies for some analytical methods, the number of significant figures to which measurements are made. The Discharger shall ensure laboratory analytical results are consistent with the requirements contained in 40 C.F.R. part 136 with regard to significant figures.

Source of Drinking Water

Any water designated as municipal or domestic supply (MUN) in the Colorado River Basin Water Board's Basin Plan.

Standard Deviation (σ)

Standard Deviation is a measure of variability that is calculated as follows:

$$\sigma = \left(\frac{\sum[(x - \mu)^2]}{(n - 1)} \right)^{0.5}$$

where:

- x is the observed value;
- μ is the arithmetic mean of the observed values; and
- n is the number of samples.

State Implementation Policy (SIP)

The Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California.

Statistic

Statistic is a computed or estimated quantity such as the mean, standard deviation, or Coefficient of Variation.

Technology-Based Effluent Limitation

A technology-based effluent limitation is a permit limit for a pollutant that is based on the capability of a treatment method to reduce the pollutant to a certain concentration.

Teratogenic

Teratogenic pollutants are substances that are known to cause structural abnormalities or birth defects in living organisms.

Test Acceptability Criteria (TAC)

Test acceptability criteria are test method-specific criteria for determining whether toxicity test results are acceptable. The effluent and reference toxicant must meet specific criteria as defined in the test method (e.g., for the Ceriodaphnia dubia survival and reproduction test, the criteria are as follows: the test must achieve at least 80 percent survival and an average of 15 young per surviving female in the control and at least 60% of surviving organisms must have three broods).

Total Maximum Daily Load (TMDL)

A TMDL is the sum of the individual waste load allocations and load allocations for receiving water. A margin of safety is included with the two types of allocations so that any additional loading, regardless of source, would not produce a violation of water quality standards.

Total Solids

Total Solids are the materials that remain as residue when dried at 103 to 105 degrees Celsius.

Toxicity Reduction Evaluation (TRE)

TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.

Toxicity Test

Toxicity test is a procedure to determine the toxicity of a chemical or an effluent using living organisms. A toxicity test measures the degree of effect on exposed test organisms of a specific chemical or effluent.

Treatment Works

Treatment works is either a federally owned, publicly owned, or privately owned device or system used to treat (including recycling and reclamation) either domestic sewage or a combination of domestic sewage and industrial waste of a liquid nature.

t-Test

t-Test (formally Student's t-Test) is a statistical analysis comparing two sets of replicate observations, in the case of WET, only two test concentrations (e.g., a control and IWC). The purpose of this test is to determine if the means of the two sets of observations are different (e.g., if the 100-percent effluent or ambient concentration differs from the control [i.e., the test passes or fails]). The statistical significance (i.e., pass/fail) of a two-sample test can be determined with either a standard t-test (if homogeneity of variance is achieved) or a modified t-test (if homogeneity of variance is not achieved).

Vector Attraction

Vector Attraction is the characteristic of a material that attracts rodents, flies, mosquitoes, or other organisms capable of transporting infectious agents.

Waste Load Allocation (WLA)

The portion of a receiving water's total maximum daily load that is allocated to one of its existing or future point sources of pollution.

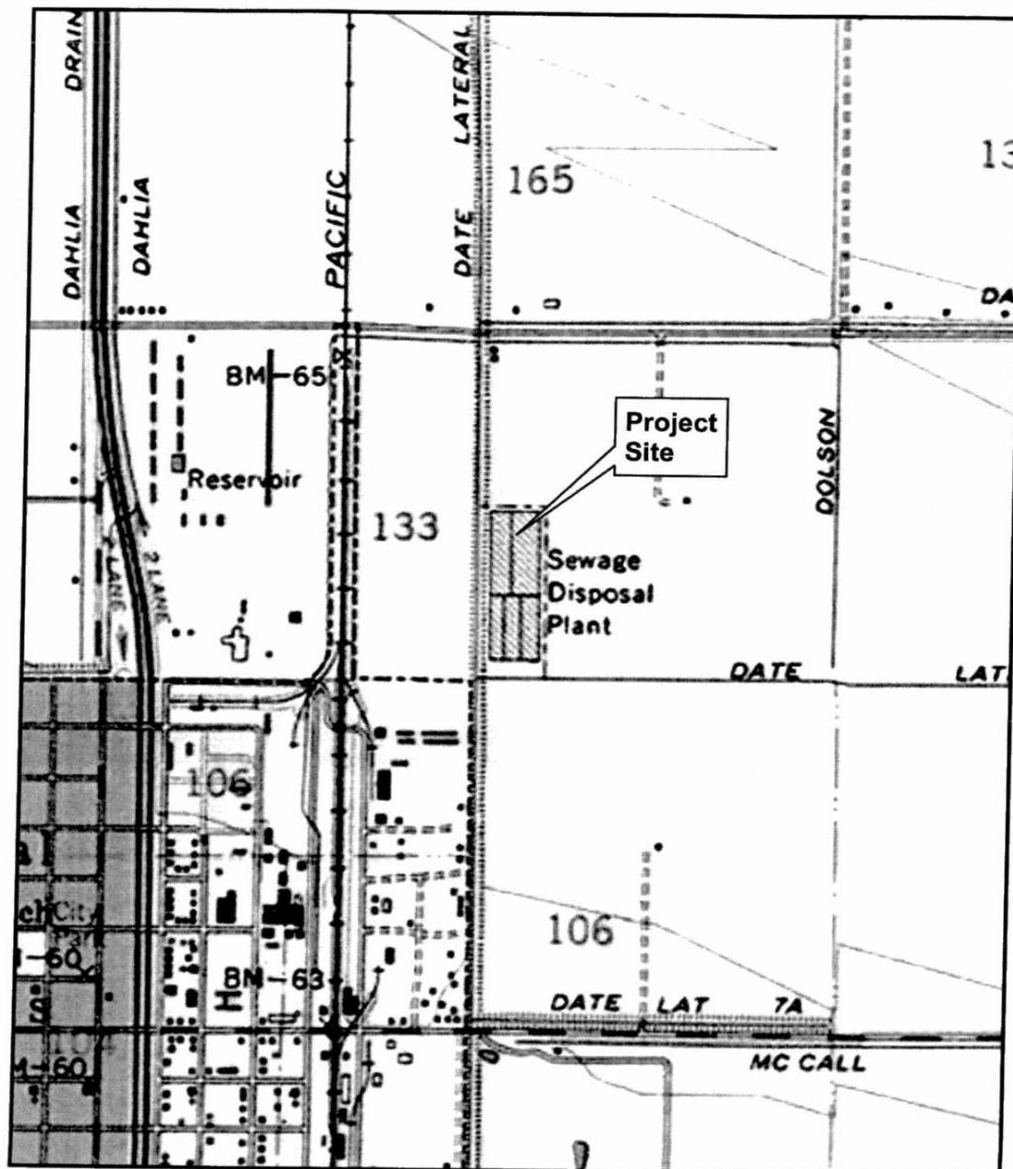
Welch's t-Test

Welch's t-Test is an adaptation of the Student's t-test intended for use with two samples having unequal variances.

Whole Effluent Toxicity (WET)

The aggregate toxic effect of an effluent measured directly by a toxicity test.

ATTACHMENT B – MAP



CITY OF IMPERIAL, OWNER/OPERATOR;
CITY OF IMPERIAL WATER POLLUTION CONTROL PLANT;
Imperial, Imperial County

Location of Discharge: Section 7, T15S, R14E, SBB&M
(Latitude 32° 51' 18" N, Longitude 115° 33' 31" W)

ATTACHMENT D – STANDARD PROVISIONS

I. STANDARD PROVISIONS – PERMIT COMPLIANCE

A. Duty to Comply

1. The Discharger must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA) and the California Water Code and is grounds for enforcement action; permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (40 C.F.R. § 122.41(a); Wat. Code §§ 13261, 13263, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
2. The Discharger shall comply with effluent standards or prohibitions established under Section 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under Section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (40 C.F.R. § 122.41(a)(1).)

B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Discharger in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 C.F.R. § 122.41(c).)

C. Duty to Mitigate

The Discharger shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 C.F.R. § 122.41(d).)

D. Proper Operation and Maintenance

The Discharger shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Discharger to achieve compliance with the conditions of this Order. Proper operation and maintenance also includes adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Discharger only when necessary to achieve compliance with the conditions of this Order. (40 C.F.R. § 122.41(e).)

E. Property Rights

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 C.F.R. § 122.41(g).)
2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 C.F.R. § 122.5(c).)

F. Inspection and Entry

The Discharger shall allow the Colorado River Basin Water Board, State Water Board, U.S. EPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (33 U.S.C. § 1318(a)(4)(B); 40 C.F.R. § 122.41(i); Wat. Code, §§ 13267, 13383):

1. Enter upon the Discharger's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 U.S.C. § 1318(a)(4)(B)(i); 40 C.F.R. § 122.41(i)(1); Wat. Code, §§ 13267, 13383);
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 U.S.C. § 1318(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(2); Wat. Code, §§ 13267, 13383);
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 U.S.C. § 1318(a)(4)(B)(ii); 40 C.F.R. § 122.41(i)(3); Wat. Code, §§ 13267, 13383); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the Water Code, any substances or parameters at any location. (33 U.S.C. § 1318(a)(4)(B); 40 C.F.R. § 122.41(i)(4); Wat. Code, §§ 13267, 13383.)

G. Bypass

1. Definitions
 - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 C.F.R. § 122.41(m)(1)(i).)
 - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 C.F.R. § 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Discharger may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 C.F.R. § 122.41(m)(2).)
3. Prohibition of bypass. Bypass is prohibited, and the Colorado River Basin Water Board may take enforcement action against a Discharger for bypass, unless (40 C.F.R. § 122.41(m)(4)(i)):
 - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 C.F.R. § 122.41(m)(4)(i)(A));
 - b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 C.F.R. § 122.41(m)(4)(i)(B)); and
 - c. The Discharger submitted notice to the Colorado River Basin Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 C.F.R. § 122.41(m)(4)(i)(C).)
4. The Colorado River Basin Water Board may approve an anticipated bypass, after considering its adverse effects, if the Colorado River Basin Water Board determines that

it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 C.F.R. § 122.41(m)(4)(ii).)

5. Notice

- a. Anticipated bypass. If the Discharger knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. (40 C.F.R. § 122.41(m)(3)(i).)
- b. Unanticipated bypass. The Discharger shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). (40 C.F.R. § 122.41(m)(3)(ii).)

H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Discharger. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 C.F.R. § 122.41(n)(1).)

1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 C.F.R. § 122.41(n)(2).)
2. Conditions necessary for a demonstration of upset. A Discharger who wishes to establish the affirmative defense of upset shall demonstrate, through properly signed, contemporaneous operating logs or other relevant evidence that (40 C.F.R. § 122.41(n)(3)):
 - a. An upset occurred and that the Discharger can identify the cause(s) of the upset (40 C.F.R. § 122.41(n)(3)(i));
 - b. The permitted facility was, at the time, being properly operated (40 C.F.R. § 122.41(n)(3)(ii));
 - c. The Discharger submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 C.F.R. § 122.41(n)(3)(iii)); and
 - d. The Discharger complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 C.F.R. § 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Discharger seeking to establish the occurrence of an upset has the burden of proof. (40 C.F.R. § 122.41(n)(4).)

II. STANDARD PROVISIONS – PERMIT ACTION

A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Discharger for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 C.F.R. § 122.41(f).)

B. Duty to Reapply

If the Discharger wishes to continue an activity regulated by this Order after the expiration date of this Order, the Discharger must apply for and obtain a new permit. (40 C.F.R. § 122.41(b).)

C. Transfers

This Order is not transferable to any person except after notice to the Colorado River Basin Water Board. The Colorado River Basin Water Board may require modification or revocation and reissuance of the Order to change the name of the Discharger and incorporate such other requirements as may be necessary under the CWA and the Water Code. (40 C.F.R. § 122.41(l)(3); 122.61.)

III. STANDARD PROVISIONS – MONITORING

- A.** Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 C.F.R. § 122.41(j)(1).)
- B.** Monitoring results must be conducted according to test procedures approved under 40 C.F.R. part 136 for the analyses of pollutants unless another method is required under 40 C.F.R. subchapters N or O. In the case of pollutants for which there are no approved methods under 40 C.F.R. part 136 or otherwise required under 40 C.F.R. subchapters N or O, monitoring must be conducted according to a test procedure specified in this Order for such pollutants. (40 C.F.R. §§ 122.41(j)(4), 122.44(i)(1)(iv).)

IV. STANDARD PROVISIONS – RECORDS

- A.** Except for records of monitoring information required by this Order related to the Discharger's sewage sludge use and disposal activities, which shall be retained for a period of at least five years (or longer as required by 40 C.F.R. part 503), the Discharger shall retain records of all monitoring information, including all calibration and maintenance records and all original strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete the application for this Order, for a period of at least three (3) years from the date of the sample, measurement, report or application. This period may be extended by request of the Colorado River Basin Water Board Executive Officer at any time. (40 C.F.R. § 122.41(j)(2).)
- B.** Records of monitoring information shall include:
 - 1. The date, exact place, and time of sampling or measurements (40 C.F.R. § 122.41(j)(3)(i));
 - 2. The individual(s) who performed the sampling or measurements (40 C.F.R. § 122.41(j)(3)(ii));
 - 3. The date(s) analyses were performed (40 C.F.R. § 122.41(j)(3)(iii));
 - 4. The individual(s) who performed the analyses (40 C.F.R. § 122.41(j)(3)(iv));
 - 5. The analytical techniques or methods used (40 C.F.R. § 122.41(j)(3)(v)); and
 - 6. The results of such analyses. (40 C.F.R. § 122.41(j)(3)(vi).)
- C.** Claims of confidentiality for the following information will be denied (40 C.F.R. § 122.7(b)):
 - 1. The name and address of any permit applicant or Discharger (40 C.F.R. § 122.7(b)(1)); and
 - 2. Permit applications and attachments, permits and effluent data. (40 C.F.R. § 122.7(b)(2).)

V. STANDARD PROVISIONS – REPORTING

A. Duty to Provide Information

The Discharger shall furnish to the Colorado River Basin Water Board, State Water Board, or U.S. EPA within a reasonable time, any information which the Colorado River Basin Water Board, State Water Board, or U.S. EPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine compliance with this Order. Upon request, the Discharger shall also furnish to the Colorado River Basin Water Board, State Water Board, or U.S. EPA copies of records required to be kept by this Order. (40 C.F.R. § 122.41(h); Wat. Code, §§ 13267, 13383.)

B. Signatory and Certification Requirements

1. All applications, reports, or information submitted to the Colorado River Basin Water Board, State Water Board, and/or U.S. EPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 C.F.R. § 122.41(k).)
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of U.S. EPA). (40 C.F.R. § 122.22(a)(3).)
3. All reports required by this Order and other information requested by the Colorado River Basin Water Board, State Water Board, or U.S. EPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 C.F.R. § 122.22(b)(1));
 - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 C.F.R. § 122.22(b)(2)); and
 - c. The written authorization is submitted to the Colorado River Basin Water Board and State Water Board. (40 C.F.R. § 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Colorado River Basin Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 C.F.R. § 122.22(c).)
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared

under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 C.F.R. § 122.22(d).)

C. Monitoring Reports

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 C.F.R. § 122.41(l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Colorado River Basin Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. (40 C.F.R. § 122.41(l)(4)(i).)
3. If the Discharger monitors any pollutant more frequently than required by this Order using test procedures approved under 40 C.F.R. part 136, or another method required for an industry-specific waste stream under 40 C.F.R. subchapters N or O, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Colorado River Basin Water Board (40 C.F.R. § 122.41(l)(4)(ii).)
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order. (40 C.F.R. § 122.41(l)(4)(iii).)

D. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 C.F.R. § 122.41(l)(5).)

E. Twenty-Four Hour Reporting

1. The Discharger shall report any noncompliance that may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A written submission shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance. (40 C.F.R. § 122.41(l)(6)(i).)
2. The following shall be included as information that must be reported within 24 hours under this paragraph (40 C.F.R. § 122.41(l)(6)(ii)):
 - a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(A).)
 - b. Any upset that exceeds any effluent limitation in this Order. (40 C.F.R. § 122.41(l)(6)(ii)(B).)
3. The Colorado River Basin Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 C.F.R. § 122.41(l)(6)(iii).)

F. Planned Changes

The Discharger shall give notice to the Colorado River Basin Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 C.F.R. § 122.41(l)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 C.F.R. § 122.41(l)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 C.F.R. § 122.41(l)(1)(ii).)
3. The alteration or addition results in a significant change in the Discharger's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during the permit application process or not reported pursuant to an approved land application plan. (40 C.F.R. § 122.41(l)(1)(iii).)

G. Anticipated Noncompliance

The Discharger shall give advance notice to the Colorado River Basin Water Board or State Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 C.F.R. § 122.41(l)(2).)

H. Other Noncompliance

The Discharger shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. (40 C.F.R. § 122.41(l)(7).)

I. Other Information

When the Discharger becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Colorado River Basin Water Board, State Water Board, or U.S. EPA, the Discharger shall promptly submit such facts or information. (40 C.F.R. § 122.41(l)(8).)

VI. STANDARD PROVISIONS – ENFORCEMENT

- A.** The Colorado River Basin Water Board is authorized to enforce the terms of this permit under several provisions of the Water Code, including, but not limited to, sections 13268, 13385, 13386, and 13387.

VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

A. Publicly-Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Colorado River Basin Water Board of the following (40 C.F.R. § 122.42(b)):

1. Any new introduction of pollutants into the POTW from an indirect discharger that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 C.F.R. § 122.42(b)(1)); and

2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 C.F.R. § 122.42(b)(2).)
3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 C.F.R. § 122.42(b)(3).)

ATTACHMENT E – MONITORING AND REPORTING PROGRAM

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP)

Section 308 of the federal Clean Water Act (CWA) and sections 122.41(h), (j)-(l), 122.44(i), and 122.48 of title 40 of the Code of Federal Regulations (40 C.F.R.) require that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Colorado River Basin Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. This MRP establishes monitoring, reporting, and recordkeeping requirements that implement the federal and California laws and/or regulations.

I. GENERAL MONITORING PROVISIONS

- A.** Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge. All samples shall be taken at the monitoring locations specified below and, unless otherwise specified, before the monitored flow joins or is diluted by any other waste stream, body of water, or substance. Monitoring locations shall not be changed without notification to and the approval of the Colorado River Basin Water Board.
- B.** Appropriate flow measurement devices and methods consistent with accepted scientific practices shall be selected and used to ensure the accuracy and reliability of measurements of the volume of monitored discharges. The devices shall be installed, calibrated and maintained to ensure that the accuracy of the measurements is consistent with the accepted capability of that type of device. Devices selected shall be capable of measuring flows with a maximum deviation of less than ± 10 percent from true discharge rates throughout the range of expected discharge volumes.
- C.** All flow measurement devices shall be calibrated at least once per year or more frequently, to ensure continued accuracy of the devices.
- D.** All analyses shall be conducted at a laboratory certified for such analyses by the State Water Resources Control Board, unless otherwise specified by this Order or Monitoring and Reporting Program. Laboratories analyzing monitoring samples shall be certified by the State Water Resources Control Board, in accordance with the provision of Water Code section 13176, and must include quality assurance/quality control data with their reports.
- E.** The collection, preservation and holding times of all samples shall be in accordance with the test procedures under 40 C.F.R. part 136 (revised as of May 14, 1999) "Guidelines Establishing Test Procedures for the Analysis of Pollutants," promulgated by the United States Environmental Protection Agency (U.S. EPA), unless otherwise specified in this MRP. In addition, the Colorado River Basin Water Board and/or EPA, at their discretion, may specify test methods that are more sensitive than those specified in 40 C.F.R part 136.
- F.** The permittee must utilize analytical methods as follows:
 - 1. A test procedure listed in 40 C.F.R. part 136.3; or
 - 2. An alternative test procedure approved by EPA as provided in 40 C.F.R. parts 136.4 or 136.5; or;
 - 3. A test procedure listed in 40 C.F.R. part 136, with modifications allowed by EPA as provided in 40 C.F.R. section 136.6.

Guidance on procedures for approval of alternative and new test procedures can be obtained from the following references: Protocol for EPA Approval of Alternative Test Procedures for Organic and Inorganic Analytes in Wastewater and Drinking Water (EPA 821-B-98-002, March 1999); and Protocol for EPA Approval of New Methods for Organic and Inorganic Analytes in Wastewater and Drinking Water (EPA 821-B-98-003, March 1999).

- G.** For priority pollutants, the Discharger shall require its testing laboratory to calibrate the analytical system down to the minimum levels (MLs) specified in 40 C.F.R. part 136, unless

an alternative minimum level is approved by the Colorado River Basin Water Board's Executive Officer. For priority pollutants with water quality-based effluent limitations (WQBELs) established in this Order, when there is more than one ML value listed in 40 C.F.R. part 136 for that substance, the Discharger shall select any one of the ML values and its associated analytical method that is below the calculated effluent limitation. If no ML is below the effluent limitation, then the lowest ML value and its associated analytical method shall be used. For priority pollutants without effluent limitations established in this Order, the Discharger shall select any one of the cited analytical methods for monitoring and reporting purposes. Any internal quality control data associated with the sample shall be reported when requested by the Executive Officer. The Colorado River Basin Water Board will reject the quantified laboratory data if quality control data is unacceptable.

- H.** In conformance with federal regulations 40 C.F.R. section 122.45(c), analyses to determine compliance with the effluent limitations for metals shall be conducted using the total recoverable method. For Chromium (VI), the dissolved method in conformance with 40 C.F.R. part 136 shall be used to measure compliance with a Chromium (VI) effluent limitation.

For Cyanide¹, analytical test methods in conformance with 40 C.F.R. part 136 shall be used as acceptable methods to measure Cyanide².

- I.** In accordance with the test procedures under 40 C.F.R. part 136, samples shall be analyzed as soon as possible after collection. The Discharger has developed a written Quality Assurance Plan (QAP) for samples that are analyzed on-site (e.g., pH³ and temperature) for the purposes of reporting compliance with effluent limitations contained in the Order. The QAP shall at a minimum address the following steps:

1. Provide a description of Standard Operating Procedures (SOPs);
2. Provide an overview of the task description and objectives;
3. Identify the sampling process, method and handling;
4. Identify the instrumentation/equipment testing, inspection and maintenance;
5. Identify the instrumentation/equipment calibration and frequency;
6. Identify the sample analysis methods and calibration range; and
7. Summarize the data review and validation procedures.

- J.** All monitoring instruments and devices used by the Discharger to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy. In the event that continuous monitoring equipment is out of service for period greater than 24-hours, the Discharger shall obtain representative grab samples each

¹ The sample for cyanide measurement shall be collected as a grab sample. Various sample preservation and sample stabilizations procedures are available that may resolve analytical interferences associated with cyanide analysis of treated wastewater effluent, ASTM Standard Practice D7365-09a. Furthermore, any technique for removal or suppression of interferences may be employed, provided the laboratory demonstrates that it more accurately measures cyanide through quality control measures described in the analytical test method. Any removal or suppression technique not described in D7365-09a or the analytical test method must be documented with supporting data.

² Federal Register, Vol. 77, No. 97, May 18, 2012. Cyanide exists in a variety of forms. It can be free or part of strong or weak complexes with other species. The analytical method employed determines what type of cyanide is measured. Types of cyanide measured include: Total, Available, Amenable to Chlorination, Weak Acid Dissociable, Free and others.

³ For pH, the Discharger may use the Myron L Ultra Meter II employing Standard Method 4500-H+B. Also, for temperature, the Discharger uses an H-B Instrument Company thermometer employing Standard Method 2550 B.

day the equipment is out of service. The Discharger shall correct the cause(s) of failure of the continuous monitoring equipment as soon as practicable. The Discharger shall report the period(s) during which the equipment was out of service and if the problem has not been corrected, shall identify the steps which the Discharger is taking or proposes to take to bring the equipment back into service and the schedule for these actions.

- K.** Monitoring results, including noncompliance, shall be reported at intervals and in a manner specified in this MRP.
- L.** Whenever the Discharger monitors any pollutant more frequently than is required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the discharge monitoring report specified by the Executive Officer.
- M.** If the facility is not in operation, or there is no discharge during a required reporting period, the Discharger shall indicate that there has been no activity during the required reporting period in CIWQS.
- N.** The Discharger shall submit values in eSMR as required to determine compliance with the permit effluent limit requirements (i.e., AMEL, MDEL, % removals, Geomeans, mass loadings, etc.).
- O.** The Discharger shall ensure that the results of the Discharge Monitoring Report-Quality Assurance (DMR-QA) Study or the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Resources Control Board at the following address:

State Water Resources Control Board Quality Assurance Program Officer
Office of Information Management and Analysis
1001 I Street, 16-39D
Sacramento, CA 95814

- P.** The Discharger shall submit the Electronic Self-Monitoring Reports (eSMRs) for reporting periods September and Third Quarter 2015 in the eSMR module for Order No. R7-2010-0020, as amended by Special Order R7-2011-0018, in the California Integrated Water Quality System (CIWQS) Program.

II. MONITORING LOCATIONS

The Discharger shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Table E-1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
--	INF-001	Wastewater influent to the treatment facility. The sampling station shall be located upstream of any in-plant return flows where a representative influent sample to the treatment plant can be obtained.
001	EFF-001	Effluent discharged from the treatment facility shall be monitored at the outlet pipe ¹ to the receiving water (Dolson Drain) where representative samples of the effluent can be obtained; Latitude 32° 51' 15" N and Longitude 115° 33' 41" W.
---	RSW-001	Receiving water (Dolson Drain) begins at the point of discharge; therefore, there is no opportunity to collect data from the receiving water from a location upstream of the discharge point.

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
---	RSW-002	Receiving water (Dolson Drain) monitoring location not to exceed 200 feet downstream from the location where the effluent enters the Dolson Drain, at a point where a plume would be expected.
---	SLD-001	Sludge quantity, location and method of disposal from the Facility.

¹ The North latitude and West longitude information in Table 1 are approximate for administrative purposes.

III. INFLUENT MONITORING REQUIREMENTS

A. Monitoring Location INF-001

- The Discharger shall monitor influent at INF-001 as follows:

Table E-2. Influent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Biochemical Oxygen Demand (BOD) (5 day @ 20 Deg. C)	mg/L	24-Hr. Composite	2x/Month	See Section I.E. of the MRP
	lbs/day	Calculated		
Suspended Solids, Total (TSS)	mg/L	24-Hr. Composite	2x/Month	"
	lbs/day	Calculated		

IV. EFFLUENT MONITORING REQUIREMENTS

A. Monitoring Location EFF-001

- The Discharger shall monitor secondary treated wastewater effluent at Monitoring Location EFF-001 as follows. If more than one analytical test method is listed for a given parameter, the Discharger must select from the listed methods and corresponding Minimum Level:

Table E-3. Effluent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	MGD	Recorder / Flow Meter Reading	1x/Day	See Section I.B of the MRP
pH	Standard Units	Grab	1x/Day	See Section I.E and I.F of the MRP
Escherichia coli (<i>E. coli</i>)	MPN/100 ml	Grab	5x/Month ¹	²
Fecal coliform	MPN/100 ml	Grab	5x/Month ¹	See Section I.E and I.F of the MRP
Temperature	°F	Grab	1x/Week	"
TSS	mg/L	24-Hr. Composite ³	2x/Month	"
	lbs/day	Calculated		
20°C BOD ₅	mg/L	24-Hr. Composite	2x/Month	"
	lbs/day	Calculated		
Copper, Total Recoverable	µg/L	Grab	1x/Month	See Section I.G of the MRP
	lbs/day	Calculated		

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Dissolved Oxygen	mg/L	Grab	1x/Month	See Section I.E and I.F of the MRP
Oil and Grease, Total ⁴	mg/L	Grab	1x/Month	"
Dissolved Solids, Total	mg/L	Grab	1x/Quarter	"
Nitrates, as N	mg/L	Grab	1x/Quarter	"
Nitrites, as N	mg/L	Grab	1x/Quarter	"
Ammonia, as N	mg/L	Grab	1x/Quarter	"
Total Nitrogen, as N	mg/L	Grab	1x/Quarter	"
Total Phosphate, as P	mg/L	Grab	1x/Quarter	"
Ortho-Phosphate, as P	mg/L	Grab	1x/Quarter	"
Hardness, Total (as CaCO ₃)	mg/L	Grab	1x/Quarter	"
Priority Pollutants ⁵	µg/L	24-Hr. Composite	1x/Year	See Section I.F of the MRP

¹ Five samples equally spaced over a 30-day period with a minimum of one sample per week.

² The Discharger may monitor for *E. coli* using analytical methods, Standard Method 9221.F or 9223 (APHA. 1998, 1995, 1992. Standard Methods for the Examination of Water and Wastewater. American Public Health Association, 20th, 19th, and 18th Editions. Amer. Publ. Hlth. Assoc., Washington D.C).

³ 24-hour composite samples shall be time-proportionate composite samples.

⁴ Total oil and grease shall include the polar and non-polar fraction of oil and grease materials.

⁵ All Priority Pollutants as defined by the California Toxics Rule (CTR), codified at 40 C.F.R. section 131.38.

V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

A. Monitoring Requirements

1. Toxicity tests shall be performed to evaluate the toxicity of the discharged wastewater in accordance with the following procedures unless otherwise specified by the Colorado River Basin Water Board's Executive Officer or his designee.

- a. Freshwater Species and Test Methods for the Chronic Test:

The toxicity tests shall be conducted in accordance with the protocol given in EPA/821-R-02-013 – Short Term Methods for Estimating the Chronic Toxicity of Effluent and Receiving Waters to Freshwater Organisms, 4th Edition.

The permittee shall conduct static renewal toxicity tests, with the fathead minnow (*Pimephales promelas*), (Larval Survival and Growth Test Method 1000.0) and the water flea (*Ceriodaphnia dubia*), (Survival and Reproduction Test Method 1002.0); and static tests with the green alga (*Selenastrum capricornutum*), (Growth Test Method 1003.0). See Table E-4 for toxicity tests.

2. The Discharger shall conduct chronic toxicity testing on the final effluent measured at Monitoring Location EFF-001 as follows:

Table E-4. Whole Effluent Toxicity Test Species

Test (s)	Species	Endpoints	Test Duration (days)	References	Sample Type	Minimum Sampling Frequency ⁴
Chronic	Fathead Minnow (<i>Pimephales promelas</i>) ¹	Larval Survival and Growth	7	EPA 821-R-02-013 (Chronic) EPA Method 1000.0	24-Hr. Composite	1x/Year ²
Chronic	Water Flea (<i>Ceriodaphnia dubia</i>) ¹	Survival and Reproduction	6-8 ³	EPA 821-R-02-013 (Chronic) EPA Method 1002.0	24-Hr. Composite	1x/Year ²
Chronic	Green Alga (<i>Selenastrum capricornutum</i>) ¹	Growth	4	EPA 821-R-02-013 (Chronic) EPA Method 1003.0	24-Hr. Composite	1x/Year ²

¹ For the fathead minnow and the water flea, the sample should consist of three samples collected on three separate days as noted in the method. The green algae test uses only one sample, as it is a shorter test.

² The screening phase (conducted during the first and fourth years of the permit term) shall be completed after a minimum of one (1) toxicity test has been completed on the three test species. The monitoring phase shall be conducted after the initial screening and during the remaining years (i.e., second, third, and fifth years of the permit term), using the most sensitive species.

³ Test duration is determined by production of 3rd brood by control and can be between 6 and 8 days.

⁴ The reporting period will match the sampling frequency (e.g., minimum sampling frequency is monthly then the reporting period is monthly, minimum sampling frequency is quarterly then reporting period is quarterly, etc.)

3. During the first and fourth years of the permit term, the toxicity testing shall be conducted in two phases, the screening phase and the monitoring phase.
 - a. For the screening phase, the Discharger shall split a 24-hour composite effluent sample and conduct concurrent toxicity tests using a fish, an invertebrate and an aquatic plant species. The fathead minnow (*Pimephales promelas*), water flea (*Ceriodaphnia dubia*), and green alga (*Selenastrum capricornutum*) are the test species approved by the Colorado River Basin Water Board's Executive Officer. The screening phase shall be completed after a minimum of one (1) toxicity test has been completed on the three test species.
 - b. For the monitoring phase, toxicity testing shall be conducted on the most sensitive species. The most sensitive species shall be selected based on the most sensitive endpoint (i.e., lethal or sub-lethal) from chronic tests conducted during the screening phase. The most sensitive species is the fish, invertebrate, or alga species which consistently demonstrates the largest percent effect level among all test endpoints at the In-stream Waste Concentration⁴ (IWC), where: IWC percent effect level = $[(\text{Control mean response} - \text{IWC mean response}) \div \text{Control mean response}] \times 100$. After the screening phase, the permittee shall then continue to conduct routine semiannual toxicity testing using the single, most sensitive species for until the next screening phase. An example of a sensitivity comparison is shown in Table E-5.

⁴ Mixing zones or dilution credits are not authorized for this discharger and 100% effluent will be considered the IWC.

Table E-5. Example of Screening Table for Chronic Test

Species	Endpoints	Mean Control Response	Mean Response at IWC (100% effluent)	% effect at IWC (100% effluent)	Most Sensitive Species
Fathead Minnow	Larval Survival	10	10	$(10 - 10)/10 \times 100 = 0\%$	
Fathead Minnow	Growth	0.41	0.363	$(0.41 - 0.363)/.41 \times 100 = 11.5\%$	
Water Flea	Survival	10	9	$(10 - 9)/10 \times 100 = 10\%$	
Water Flea ¹	Reproduction	33.4	26.7	$(33.4 - 26.7)/33.4 \times 100 = 20\%$	Highest % effect represents most sensitive species ¹
Green Alga	Growth	197.3	170.1	$(197.3 - 170.1)/197.3 \times 100 = 13.8\%$	

¹ In this example, the water flea represents the most sensitive species. Chronic tests for the water flea shall be conducted as required by measuring and reporting the endpoints for survival and reproduction during the monitoring phase.

B. Quality Assurance

1. Quality assurance measures, instructions, and other recommendations and requirements are found in the chronic test methods manual previously referenced. Additional requirements are specified below.
2. Control water should be prepared and used as specified in the test method manual Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms (EPA/821/R-02/013, 2002). Dilution and control waters may be obtained from an unaffected area of receiving waters. Synthetic (standard) dilution is an option and may be used if the above source is suspected to have toxicity greater than 1.0 TUC.
3. A series of at least five dilutions and a control shall be tested for chronic toxicity testing if not using the t-test or modified t-test. The series shall include the following concentrations: 12.5, 25, 50, 75, and 100 percent effluent.
4. For the chronic toxicity testing using a t-test, two dilutions shall be used, i.e., 100 percent effluent and a control. The statistical significance (i.e., pass/fail) of a two-sample test can be determined with either a standard t-test (if homogeneity of variance is achieved) or a modified t-test (if homogeneity of variance is not achieved).
5. If organisms are not cultured in-house, testing laboratories shall conduct concurrent testing with a reference toxicant. If organisms are cultured in-house, then monthly reference toxicant testing is sufficient. Reference toxicant tests shall also be conducted using the same test conditions as the effluent toxicity tests (e.g., same test duration, etc.). Testing laboratories shall perform a reference toxicant test quarterly, concurrently with each effluent toxicity test. Reference toxicant testing is used to document ongoing laboratory performance in addition to assessing the sensitivity of the test organism.

6. All reference toxicant test results must be reviewed and reported according to EPA guidance on the evaluation of concentration-response relationships found in Method Guidance and Recommendations for Whole Effluent Toxicity (WET) Testing (40 C.F.R. part 136) (EPA 821-B-00-004, 2000).
7. If either the reference toxicant test or effluent test does not meet all test acceptability criteria (TAC) as specified in the toxicity test references, then the Discharger must re-sample and retest within 15 working days or as soon as possible. The retesting period begins when the Discharger receives the test results that indicate retesting is needed.
8. The reference toxicant and effluent tests must meet the upper and lower bounds on test sensitivity as determined by calculating the percent minimum significant difference (PMSD) for each test result. The test sensitivity bound is specified for each test method in the respective methods manuals.
9. If the discharged effluent is chlorinated, then chlorine shall not be removed from the effluent sample prior to toxicity testing without written approval by the permitting authority.
10. A pH drift during a toxicity test may contribute to artifact toxicity when pH-dependent toxicants (e.g., ammonia, metals) are present in the effluent. To determine whether or not pH drift is contributing to artifact toxicity, the permittee shall conduct three sets of side-by-side toxicity tests in which the pH of one treatment is controlled at the pH of the effluent while the pH of the other treatment is not controlled, as described in Section 11.3.6.1 of Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms (EPA/821/R-02/013, 2002). Toxicity is confirmed to be an artifact and due to pH drift when no toxicity above the chronic WET permit limit or trigger is observed in the treatments controlled at the pH of the effluent. Upon this confirmation, the permittee shall request and upon written approval by the Colorado River Basin Water Board's Executive Officer, the permittee may use the procedures outlined in Section 11.3.6.2 of the chronic freshwater test methods manual to control effluent sample pH during the toxicity test.

C. Chronic Toxicity Definition and Numeric Toxicity Whole Effluent Toxicity (WET) Monitoring Triggers

1. Chronic Toxicity Definition.
 - a. Chronic toxicity measures sub-lethal effect (e.g., reduced growth, reproduction) to experimental test organisms exposed to an effluent or ambient waters compared to that of the control organisms.
 - b. Chronic toxicity shall be measured in TU_c , where $TU_c = 100/NOEC$. The no observed effect concentration (NOEC) is the highest concentration of toxicant to which organisms are exposed in a chronic test that causes no observable adverse effect on the test organisms (e.g., the highest concentration of toxicant to which the values for the observed responses are not statistically significantly different from the control(s)).
 - c. If using a t-test or modified t-test, chronic toxicity shall be reported as pass/fail using a laboratory control and the sample (e.g., 100% effluent) during the test. The determination of pass or fail from a single aqueous concentration is ascertained with a standard t-test (refer to Appendix H of EPA's Short Term Methods for Estimating the Chronic Toxicity of Effluent and Receiving Waters to Freshwater Organisms, 4th Edition (EPA/821-R-02-013). In these pass/fail tests, the objective is to determine if the survival in the single treatment (e.g., effluent) is significantly different from the

control survival. EPA Region 9 recommends the statistical significance (i.e., pass/fail) of a two-sample test design be determined with either a modified t-test (if homogeneity of variance is not achieved) or a standard t-test (if homogeneity of variance is achieved).

2. Numeric Chronic Toxicity Monitoring Trigger.
 - a. Any chronic toxicity test result that exceeds 1.6 chronic toxicity units (TU_c) or a monthly median (consecutive samples) that exceeds 1 TU_c shall trigger accelerated WET testing.
 - b. Any chronic toxicity test result that results in "fail" when using a t-test or modified t-test shall trigger accelerated monitoring.

D. Accelerated Toxicity Testing and TRE/TIE Process

1. If the chronic WET permit trigger is exceeded and the source of toxicity is known [e.g., a temporary plant upset, ammonia, ionic imbalance or elevated total dissolved solids (TDS)], then the permittee shall conduct one additional toxicity test. The permittee shall use the same species and test method that failed the WET test. This toxicity test shall begin within 14 days of receipt of a test result exceeding the chronic WET permit trigger. If the additional toxicity test does not exceed the WET permit trigger or it is confirmed that the toxicity is due to temporary plant upset, ammonia, ionic imbalance or elevated TDS, then the permittee may return to the regular testing frequency.
2. If the chronic WET permit trigger is exceeded and the source of toxicity is not known, then the permittee shall conduct three additional toxicity tests using the same species and test method, approximately every two weeks, over a 6-week period. This testing shall begin within 14 days of receipt of a test result exceeding the chronic WET permit trigger. If none of the additional toxicity tests exceed the chronic WET permit trigger, then the permittee may return to the regular testing frequency.
3. If one of the additional toxicity tests, in paragraphs V.D.1 and V.D.2 above, exceeds the chronic WET permit trigger, then, within 14 days of receipt of this test result, the permittee shall initiate a TIE.
4. The permittee may initiate a TIE as part of a TRE to identify the causes of toxicity using the same species and test method. The TIE shall be conducted to identify and evaluate toxicity in accordance with procedures recommended by the U.S. EPA which include the following:
 - a. Toxicity Identification Evaluations: Characterization of Chronically Toxic Effluents, Phase I (EPA/600/6-91/005F, 1992);
 - b. Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures (EPA/600/6-91/003, 1991);
 - c. Methods for Aquatic Toxicity Identification Evaluations, Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity (EPA/600/R-92/080, 1993); and
 - d. Methods for Aquatic Toxicity Identification Evaluations, Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity (EPA/600/R-92/081, 1993).
5. As part of the TIE Investigation, the Discharger shall be required to implement its TRE work plan. The TRE Work Plan which shall include the following: further actions undertaken by the permittee to investigate, identify, and correct the causes of toxicity; actions the permittee will take to mitigate the effects of the discharge and prevent the

recurrence of toxicity; and a schedule for these actions. The Discharger shall take all reasonable steps to control toxicity once the source of the toxicity is identified. A failure to conduct required WET tests or a TRE within a designated period shall result in the establishment of numerical toxicity effluent limitations in a permit or appropriate enforcement action. Recommended guidance in conducting a TRE includes the following:

- a. Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants, EPA/833B-99-002, August 1999;
- b. Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations, EPA/600/2-88/70, April 1989; and
- c. Clarifications Regarding Toxicity Reduction and Identification Evaluations in the National Pollutant Discharge Elimination System Program dated March 27, 2001, U.S. EPA Office of Wastewater Management, Office of Regulatory Enforcement.

E. Ammonia, Ionic Imbalance or Elevated TDS Toxicity

1. For discharges where a TIE has identified ammonia as a cause of toxicity, the permittee shall calculate the response threshold on the basis of unionized and total ammonia. The permittee shall run a parallel test with ammonia in lab water to evaluate if the lab water and the effluent responses are the same (i.e., no matrix effect). In future WET testing, where ammonia toxicity is hypothesized as the cause, the permittee has the following three options to evaluate whether ammonia is causing the toxicity:
 - a. If toxicity in lab water is similar to that in the effluent, the permittee shall conduct a parallel test with ammonia spiked into lab water. Toxicity endpoints are compared on the basis of unionized ammonia. If the endpoints are the same, then the implication is ammonia is responsible for toxicity and no further action is required; or
 - b. If toxicity in lab water is not similar to that in the effluent, the permittee shall conduct a parallel test with effluent, maintaining pH at a level that maintains the unionized fraction below the toxic threshold. If no toxicity is observed in the pH controlled sample, then implication is that ammonia is responsible for toxicity and no further action is required; or
 - c. Without using comparative tests, calculate toxicity in the sample on the basis of unionized ammonia and compare the result to data generated in the TIE; if the results support the hypothesis that ammonia explains toxicity, then no further action is required.

Using these approaches, if ammonia is identified as the toxicant, the permittee shall document the results and findings in the monitoring report and no further testing is required. However, if ammonia is not identified as the toxicant, the permittee shall take action as described in Section D. Accelerated Toxicity Testing and TRE/TIE Process of this permit.

2. For discharges where a TIE has identified ionic imbalance or elevated TDS as a cause of toxicity, the permittee shall conduct the following concurrent tests to characterize the contribution of ionic imbalance or elevated TDS to effluent toxicity. Based on the results from the TIE, toxicity should be either quantitatively recovered in synthetic effluent that mimics ionic imbalance or elevated TDS, or eliminated by adding selected ions to the effluent to address deficiencies. Thus, in future WET testing, where ionic imbalance or elevated TDS is hypothesized as contributing to toxicity, the permittee has the following two options to evaluate whether ionic imbalance or elevated TDS is causing the toxicity:

- a. Conducting a parallel test with synthetic effluent that mimics the ionic imbalance or TDS concentration; or
- b. Conducting a parallel test with effluent spiked with deficient ion(s).

Using these approaches, if ionic imbalance or elevated TDS is shown to account for toxicity, the permittee shall document the results and findings in the monitoring report and no further testing is required. However, if the parallel tests do not account for toxicity, the permittee shall take action as described in Section D. Accelerated Toxicity Testing and TRE/TIE Process of this permit.

F. Reporting of Toxicity Monitoring Results

1. The permittee shall submit either a summary page or the full laboratory report for all toxicity testing as an attachment to CIWQS for the reporting period (e.g., monthly, quarterly, semi-annually or annually) and provide the data (i.e., TUc, TUa or Pass/Fail) in the PET tool for uploading into CIWQS. The laboratory report shall contain: the toxicity test results (TUc or pass/fail and percent effect); the dates of sample collection and initiation of each toxicity test; all results for effluent parameters monitored concurrently with the toxicity test(s); and progress reports on TRE/TIE investigations.
2. The permittee shall provide the actual test endpoint responses for the control (i.e., the control mean) and the IWC (i.e., the IWC mean) for each toxicity test to facilitate the review of test results and determination of reasonable potential for chronic WET by the permitting authority.
3. The permittee shall notify the permitting authority in writing within 14 days of exceedance of the chronic WET permit trigger. This notification shall describe actions the permittee has taken or will take to investigate, identify, and correct the causes of toxicity; the status of actions required by this permit; and schedule for actions not yet completed; or reason(s) that no action has been taken.

VI. LAND DISCHARGE MONITORING REQUIREMENTS – NOT APPLICABLE

VII. RECYCLING MONITORING REQUIREMENTS – NOT APPLICABLE

VIII. RECEIVING WATER MONITORING REQUIREMENTS

A. Monitoring Location RSW-001

1. The Discharger shall monitor the Dolson Drain at RSW-001 as follows. In the event that no receiving water is present at RSW-001, no receiving water monitoring data are required for station RSW-001.

Table E-6. Receiving Water Monitoring Requirements – RSW-001

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
pH	Standard Units	Grab	1x/Quarter	See Section I.E of the MRP
Temperature	°F	Grab	1x/Quarter	"
Dissolved Oxygen	mg/L	Grab	1x/Quarter	"
Total Hardness (as CaCO ₃)	mg/L	Grab	1x/Quarter	"
Total Dissolved Solids	mg/L	Grab	1x/Quarter	"
Nitrates, as N	mg/L	Grab	1x/Quarter	"
Nitrites, as N	mg/L	Grab	1x/Quarter	"
Ammonia, as N	mg/L	Grab	1x/Quarter	"

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Total Nitrogen, as N	mg/L	Grab	1x/Quarter	"
Total Phosphate, as P	mg/L	Grab	1x/Quarter	"
Ortho-Phosphate, as P	mg/L	Grab	1x/Quarter	"
Priority Pollutants ¹	µg/L	Grab	1x/Year	See Section I.F of the MRP

¹ All Priority Pollutants as defined by the California Toxics Rule (CTR), codified at 40 C.F.R. section 131.38.

B. Monitoring Location RSW-002

- The Discharger shall monitor the Dolson Drain at RSW-002 as follows. In the event that no receiving water is present at RSW-002, no receiving water monitoring data are required for station RSW-002.

Table E-7. Receiving Water Monitoring Requirements- RSW-002

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
pH	Standard Units	Grab	1x/Quarter	See Section I.E of the MRP
Temperature	°F	Grab	1x/Quarter	"
Dissolved Oxygen	mg/L	Grab	1x/Quarter	"
Total Hardness (as CaCO ₃)	mg/L	Grab	1x/Quarter	"
Total Dissolved Solids	mg/L	Grab	1x/Quarter	"
Nitrates, as N	mg/L	Grab	1x/Quarter	"
Nitrites, as N	mg/L	Grab	1x/Quarter	"
Ammonia, as N	mg/L	Grab	1x/Quarter	"
Total Nitrogen as N	mg/L	Grab	1x/Quarter	"
Total Phosphate, as P	mg/L	Grab	1x/Quarter	"
Ortho-Phosphate, as P	mg/L	Grab	1x/Quarter	"

C. Visual Monitoring

- In conducting the receiving water sampling, a log shall be kept of the receiving water conditions at Monitoring Locations RSW-001 and RSW-002. Notes on receiving water conditions shall be summarized in the monthly monitoring report and when data are submitted electronically via the SMR module in the CIWQS Program, data shall be reported in the "Attachments" section. Attention shall be given to the presence or absence of:
 - Floating or suspended matter;
 - Discoloration;
 - Aquatic life (including plants, fish, shellfish, birds);
 - Visible film, sheen, or coating;
 - Fungi, slime, or objectionable growths; and
 - Potential nuisance conditions.

IX. OTHER MONITORING REQUIREMENTS

A. Monitoring Location SLD-001 Sludge Monitoring

1. In the event that sludge is generated at the treatment facility, sludge shall be sampled and analyzed for the following prior to disposal:

Table E-8. Sludge Monitoring Required SLD - 001

Parameter	Units	Sample Type ¹	Required Analytical Test Method
Arsenic	mg/kg	Composite	See Footnotes 2 and 3
Cadmium	mg/kg	Composite	"
Copper	mg/kg	Composite	"
Lead	mg/kg	Composite	"
Mercury	mg/kg	Composite	"
Molybdenum	mg/kg	Composite	"
Nickel	mg/kg	Composite	"
Selenium	mg/kg	Composite	"
Zinc	mg/kg	Composite	"
Kjeldahl Nitrogen (TKN), (as N)	mg/kg	Composite	"
Ammonia, as N	mg/kg	Composite	"
Nitrate, as N	mg/kg	Composite	"
Total Phosphorus, as P	mg/kg	Composite	"
Potassium	mg/kg	Composite	"
Total Solids	mg/kg	Composite	"
Fecal Coliform	MPN/gr am	Composite	"
Total Petroleum Hydrocarbons	mg/kg	Composite	"

¹ Representative samples shall be collected prior to use or disposal.

² Pollutants shall be analyzed using the analytical methods described in 40 C.F.R. part 503.8.

³ Results shall be reported on a 100% dry weight basis. Records of all analyses shall state on each page of the analysis results whether the results are expressed on an "as-is" basis or on a 100% dry weight basis.

In the event that sludge is generated at the treatment facility, sludge monitoring requirements listed in Table E-8, above, shall be sampled and analyzed according to the volume of sludge generated per year at the frequency specified below:

Table E-9. Sludge Monitoring Required SLD - 001

Volume Generated ¹ (dry metric tons per year)	Volume Generated (US short tons per year)	Minimum Sampling Frequency
Greater than zero, but less than 290	Greater than zero, but less than 320	1x/Year ^{1,2}
Equal to or greater than 290, but less than 1,500	Equal to or greater than 320, but less than 1,650	1x/Quarter ^{1,2}
Equal to or greater than 1,500, but less than 15,000	Equal to or greater than 1,650, but less than 16,500	6x/Year ^{1,2}
Equal to or greater than 15,000	Equal to or greater than 16,500	1x/Month ^{1,2}

¹ If sludge is removed for use or disposal on a routine basis (e.g., daily, weekly, quarterly, etc.), sampling should be scheduled at regular intervals throughout the year in accordance with Table E-9. If sludge is stockpiled at the treatment facility and is not removed for use or disposal within the applicable monitoring frequencies listed

in Table E-9, the Discharger may collect representative samples of the sludge generated at the treatment facility with the sampling frequency listed in Table E-9 or representative composite samples may be taken from the stockpile(s) prior to use or disposal. For dredging operations where sludge will be dewatered at the treatment facility, the Discharger shall collect representative composite samples from the dewatered sludge prior to use or disposal.

² The Discharger shall submit the results of the sludge monitoring requirements listed in Tables E-8 and E-9 in an annual sewage sludge monitoring report.

2. In addition to the sludge monitoring requirements listed above in Tables E-8 and E-9, the Discharger shall also include the following information in the sewage sludge monitoring report:
 - a. The volume of sludge generated at the treatment facility that year, in dry metric tons, and the amount of sludge stockpiled from previous years.
 - b. The names, mailing addresses, and street addresses of persons who received sludge generated from the treatment facility. The Discharger shall specify the volume of sludge delivered and specify if the transferred sludge is to be stored, treated, placed in surface disposal sites, land applied, incinerated, disposed in municipal solid waste landfills or disposed by some other method.
 - c. For sewage sludge to be disposed in a municipal solid waste landfill, the Discharger shall certify that the sludge does not contain "free liquids" as defined by Method 9095B (Paint Filter Liquids Test), included in "Test Methods for Evaluating Solid Waste, Physical/Chemical Methods" (EPA Publication SW-846). The Discharger shall test the sludge using the Paint Filter Liquids Test at the frequency in Table E-9, or more frequently if necessary, to demonstrate that there are no free liquids.
 - d. For sewage sludge to be applied to the land or placed on a surface disposal site, the Discharger provide the following certification:
 - i. Prior to land application, the Discharger shall demonstrate that the sewage sludge achieved the operational standards for pathogen reduction levels and vector attraction reduction requirements as required in 40 C.F.R. section 503.17 and section 503.15.
 - ii. Prior to disposal in a surface disposal site, the Discharger shall demonstrate that the sewage sludge achieved the operational standards for pathogen reduction levels and vector attraction reduction requirements as required in section 503.27 and section 503.25.
3. In accordance with Special Provision VI.C.5.a., Limitations and Discharge Requirements, the Discharger shall prepare a plan in which the methods of treatment, handling, storage, and disposal of sludge are described. Further, the Discharger shall maintain a copy of the solids management plan on-site and have available for review during inspection.

B. Pretreatment Monitoring

In the event that significant industrial wastewater are being discharged to the wastewater treatment facility, then the Discharger shall provide the Colorado River Basin Water Board with an annual report describing the pretreatment program activities over the previous twelve (12) month period and it shall include:

1. A summary of actions taken by the Discharger which ensures industrial-user compliance;
2. An updated list of industrial users (by SIC categories) which were issued permits, and/or enforcement orders; and
3. The name and address of each user that received a revised discharge limit.

In the event that an approved pretreatment program is required, the discharge shall submit a pretreatment program submittal to obtain pretreatment approval.

X. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

The Discharger shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.

1. The Discharger shall report the results of chronic toxicity testing, TRE, and TIE as required in section V, "Effluent Toxicity Testing".
2. The results of any analysis taken more frequently than required using analytical methods, monitoring procedures and performed at the locations specified in this MRP shall be reported to the Colorado River Basin Water Board.
3. The Discharger shall ensure laboratory analytical results are consistent with the requirements contained in 40 C.F.R. part 136 with regard to significant figures. 40 C.F.R. part 136 specifies for some analytical methods, the number of significant figures to which measurements are made.

B. Electronic Self-Monitoring Reports (eSMRs)

1. The Discharger shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). The CIWQS Web site will provide additional information for eSMR submittal in the event there will be a planned service interruption for electronic submittal.
2. The Discharger shall maintain sufficient staffing and resources to ensure it submits eSMRs for the duration of the term of this permit including any administrative extensions. This includes provision of training and supervision of individuals (e.g., Discharger personnel or consultant) on how to prepare and submit eSMRs.
3. The Discharger shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. The Discharger shall submit monthly, quarterly, and annual eSMRs including the results of all required monitoring using U.S. EPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last eSMR was submitted. If the Discharger monitors any pollutant more frequently than required by this Order, the results of this monitoring shall be included in the calculations and reporting of the data submitted in the eSMR.
4. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table E-10. Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Daily 1x/Day	October 1, 2015	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	Submit with monthly eSMR
Weekly 1x/Week	October 1, 2015	Sunday through Saturday	Submit with monthly eSMR
Monthly 1x/Month	October 1, 2015	1 st day of calendar month through last day of calendar month	First day of second month from end of

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
			monitoring period
Quarterly 1x/Quarter 4x/Year	October 1, 2015	January 1 through March 31 April 1 through June 30 July 1 through September 30 October 1 through December 31	May 1 August 1 November 1 February 1
Annually 1x/Year	October 1, 2015	January 1 through December 31	March 1

5. Reporting Protocols. The Discharger shall follow the procedure in 40 CFR part 136 when reporting the results of analytical determinations of chemical constituents in a sample. Further, the Discharger shall use the following reporting protocol:
 - a. Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample). For reporting concentration and calculated values in the pet tool follow these instructions:

Reporting Concentration - Under the "Qualifier" column select "=" and under the "Result" column report the result (concentration).

Reporting Calculated Values - Under the "Qualifier" column select "=" and under the "Result" column report the result (calculated value).
 - b. Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported under the "Qualifier" column as "DNQ" (Detected, but Not Quantified). For the purposes of data collection, the laboratory shall write the estimated chemical concentration under the "Result" column next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory. For reporting concentration and calculated values in the pet tool follow these instructions:

Reporting Concentration – Under the "Qualifier" column select "DNQ", under the "Result" column report the estimated chemical concentration. In addition, the MDL shall be reported under the "MDL" column and the ML shall be reported under the "ML" column.

Reporting Calculated Values – Under the "Qualifier" column select "<", under the "Result" column report the calculated value or in the case of mass loading report the average monthly effluent limitation for mass loading.
 - c. Sample results less than the laboratory's MDL shall be reported as "ND" (Not Detected). For reporting concentration and calculated values in the pet tool follow these instructions:

Reporting Concentration – Under the "Qualifier" column select "ND" and report the MDL under the "MDL" column.

Reporting Calculated Values - Under the "Qualifier" column select "<", under the "Result" column report the calculated value (Flow, mgd x 8.34 x MDL (use correct units)) or in the case of mass loading report the average monthly effluent limitation for mass loading.
 - d. Dischargers are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to

calibration standards) is the lowest calibration standard. At no time is the Discharger to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.

6. Compliance Determination. Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined above, Attachment E and section VII. Compliance Determination. For purposes of reporting and administrative enforcement by the Colorado River Basin Water Board and State Water Board, the Discharger shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).
7. Multiple Sample Data. When determining compliance with an AMEL for priority pollutants and more than one sample result is available, the Discharger shall compute the arithmetic mean unless the data set contains one or more reported determinations of "Detected, but Not Quantified" (DNQ) or "Not Detected" (ND). In those cases, the Discharger shall compute the median in place of the arithmetic mean in accordance with the following procedure:
 - a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
8. The Discharger shall submit eSMRs in accordance with the following requirements:
 - a. The Discharger shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Discharger is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When CIWQS does not provide for entry into a tabular format within the system, the Discharger shall electronically submit the data in a tabular format as an attachment.
 - b. The Discharger shall attach a cover letter to the eSMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation. In addition, the Discharger shall add these violations into CIWQS.
 - c. The Discharger shall upload the Whole Effluent Toxicity Test result page or entire report for the reporting period under the attachment tab for the reporting period.
 - d. The Discharger shall upload the laboratory reports for the analysis of the priority pollutant for the reporting period under the attachment tab for the reporting period. The Discharger shall evaluate the results with the criteria and notify the Colorado River Basin Regional Board of any exceedance of the criteria.

C. Discharge Monitoring Reports (DMRs)

The Discharger shall electronically certify and submit DMRs together with SMRs using Electronic Self-Monitoring Reports module eSMR 2.5 or any upgraded version. Electronic submittal of DMRs will be in addition to electronic submittal of SMRs. Information about electronic submittal of DMRs is available at the Discharge Monitoring Report website at http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring/.

D. Other Reports

1. The Discharger shall report the results of any special studies required by Special Provisions – VI.C.2 (TRE/TIE; acute and chronic toxicity testing, Translator Study; Spill Response Plan, and Sludge Disposal and Notification Plan) of this Order. The Discharger shall report the progress in satisfaction of compliance schedule dates specified in Special Provisions – VI.C.7 of this Order. The Discharger shall submit reports with the eSMR scheduled to be submitted on or immediately following the report due date.
2. **Operations and Maintenance Report.** The Discharger shall report the following as shown in Table E-11:

Table E-11. Operations and Maintenance Report

Activity	Reporting Frequency
To inspect and document the operation and maintenance of the UV disinfection system, including but not limited to, inspection, cleaning, and bulb replacement. The Discharger shall provide a certification statement in the annual report that inspections and documentation of inspections and operations/maintenance problems have been completed.	1x/Year
To inspect and document any operation/maintenance problems by inspecting each unit process. The Discharger shall provide a certification statement in the annual report that inspections and documentation of inspections and operations/maintenance problems have been completed.	1x/Year
Calibration of flow meters and mechanical equipment shall be performed in a timely manner and documented. The Discharger shall provide a certification statement in the annual report that the calibration of flow meters and mechanical equipment has been conducted and documentation of such calibrations is maintained.	1x/Year
The Discharger shall maintain documentation of all logbooks (operation and maintenance), chain of custody sheets, laboratory and sampling activities as stated in Standard Provisions sections IV and V (Attachment D). The Discharger shall provide a certification statement in the annual report that maintenance of logbooks, chain of custody sheets, and laboratory and sampling activities as required is being implemented.	1x/Year
The Discharger shall conduct an annual review and evaluation of priority pollutant sampling results collected each year to evaluate the impact on surface water quality, and provide this evaluation in the annual report.	1x/Year
The Discharger shall provide a certification statement in the annual report that the DMR-QA Study has been submitted to the State Water Resources Control Board Quality Assurance Office and it has been received.	1x/Year

3. **Discharge Monitoring Report-Quality Assurance (DMR-QA) Study or the most recent Water Pollution Performance Evaluation Study.** The Discharger can satisfy the requirements of the DMR-QA study using one of the following two options:
 - a) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA study; or

- b) The Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from their own laboratories or their contract laboratories.

The Discharger shall submit annually the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study to the State Water Board's Quality Assurance Officer who will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to U.S. EPA's DMR-QA Coordinator and Quality Assurance Manager.

The key components of the study are listed below.

- i. No Study time frame. The Study period is the entire twelve months of each Calendar Year. However, participation earlier in the year would allow for extra time in the event that sample retesting is necessary.
- ii. Laboratories may use the same sample data that is normally generated under their certification/accreditation obligations.
- iii. Please submit electronic data files along with PDF copies from the approved Proficiency Testing (PT) vendors. The vendor will provide electronic data files (CSV) in the form of the standard EPA data file transfer protocols for uploading into the State Water Board database for this study. You are responsible for ensuring the testing data is received by the State Water Board.

The above Study will be submitted annually to the following address:

State Water Resources Control Board Quality Assurance Program Officer
Office of Information Management and Analysis
1001 I Street, 16-39D
Sacramento, CA 95814

Contact Information

Renee Spears, Senior Environmental Specialist – QA Officer
(916) 341-5583

renee.spears@waterboards.ca.gov

ATTACHMENT F – FACT SHEET

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ATTACHMENT F – FACT SHEET

As described in section II.D of this Order, the Colorado River Basin Water Board incorporates this Fact Sheet as findings of the Colorado River Basin Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for dischargers in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Discharger. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Discharger.

I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

Table F-1. Facility Information

WDID	7A130106011
Discharger	City of Imperial
Name of Facility	City of Imperial Wastewater Treatment Plant
Facility Address	701 East 14 th Street
	Imperial, CA 92251
	Imperial County
Legally Responsible Official	Jackie Loper, Public Services Director (760) 355-3336; jloper@cityofimperial.org
Facility Contact, Title and Phone	Mark Coon, Chief Wastewater Operator (760) 455-3426; mark1605ww@hotmail.com
Authorized Person to Sign and Submit Reports	Mark Coon, Chief Wastewater Operator (760) 455-3426; mark1605ww@hotmail.com Edgar Self, Public Service Manager (760) 355-2734; eself@cityofimperial.org
Mailing Address	420 South Imperial Avenue Imperial, CA 92251
Billing Address	SAME
Type of Facility	POTW
Major or Minor Facility	Major
Threat to Water Quality	2
Complexity	A
Pretreatment Program	N
Recycling Requirements	N
Facility Permitted Flow	2.4 MGD
Facility Design Flow	2.4 MGD
Watershed	West Colorado River Basin
Receiving Water	Dolson Drain, a tributary to Alamo River via Lilac Drain and Rose Drain
Receiving Water Type	Agricultural Drain

- A.** The City of Imperial (hereinafter Discharger) is the owner and operator of the City of Imperial Wastewater Treatment Plant (hereinafter Facility), a Publicly Owned Treatment Works.

For the purposes of this Order, references to the “discharger” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Discharger herein.

- B.** The Facility discharges wastewater to the Dolson Drain, a tributary to the Alamo River, a water of the United States. The Discharger was previously regulated by Order R7-2010-0020, as amended by Special Order R7-2011-0018, and National Pollutant Discharge Elimination System (NPDES) Permit No. CA0104400 which was adopted on September 16, 2010 and which expires on September 30, 2015. Attachment B provides a map of the area around the Facility. Attachment C provides a flow schematic of the Facility.

Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a watercourse, the Discharger must file a petition with the State Water Board, Division of Water Rights, and receive approval for such a change. The State Water Board retains the jurisdictional authority to enforce such requirements under Water Code section 1211.

- C.** The Discharger filed a report of waste discharge and submitted an application for reissuance of its WDRs and NPDES permit on April 3, 2015. The application was deemed complete on June 4, 2015. A site visit was conducted on July 28, 2015, to observe operations and collect additional data to develop permit limitations and requirements for waste discharge.

II. FACILITY DESCRIPTION

The Discharger owns and operates a wastewater collection, treatment and disposal system (hereinafter referred to as the Facility) and provides service to the City of Imperial, service to a population of approximately 13,000. The wastewater treatment plant has a treatment (design) capacity of 2.4 MGD and is located in Section 7, T15S, R14E, SBB&M.

A. Description of Wastewater and Biosolids Treatment or Controls

The wastewater treatment plant consists of two parallel treatment trains and a final ultraviolet (UV) disinfection system for the combined effluent.

Raw wastewater from the City of Imperial reaches the Facility via two main conveyance lines: the “N” Line and the “15th Street” line to a central pump sump. The influent pump station then transfer raw wastewater to the gravity separator, which comprises two screened horizontal troughs that are manually cleaned. Screened materials are collected and placed in a hopper for subsequent transport to landfill.

Train 1-Oxidation Ditch: Raw wastewater from the influent sump is pumped to Train 1- Oxidation Ditch (OD) system via a Parshall flow meter. Train 1- OD system comprises two oxidation ditches each followed by a round clarifier basin. The design flow for each oxidation ditch-clarifier combination is 0.7 MGD. Return activated sludge (RAS) and waste activated sludge (WAS) from the clarifiers are pumped to the OD inlet and to the sludge drying beds, respectively. Any surface scum from the clarifiers is scraped off and conveyed to a “scum pit” where the decanted liquid is pumped to the RAS line and the remaining solids are periodically “vactored” and disposed at landfill.

Train 2–Biolac™ Activated Sludge: Screened wastewater from the influent gravity separator is also conveyed to Train 2–Biolac™ Activated Sludge system via a lift station. The Train 2- Biolac™ system was constructed in the 2007-08 period. Train 2-Biolac™ system comprises an aeration basin with submerged aerators and two integral, rectangular clarifiers. The design flow for the Biolac™ system is 1.0 MGD. Return activated sludge (RAS) and waste activated sludge (WAS) from the clarifiers are pumped to the aeration basin inlet and to the sludge drying beds, respectively.

UV Disinfection System: The treated effluent from Train 1 and Train 2 is combined just before the entry point to the UV disinfection system. The UV system comprises of one channel equipped with two racks in series of Trojan 3000 Plus equipment (open channel) and is sized for a 3 MGD flow.

Disinfected wastewater is conveyed to Discharge Point 001 via an approximately 1,200 feet long open ditch to the confluence of Dolson Drain and two other field drains that connect at the same location via an irrigation drain water pump (lift) station. The Dolson Drain is a tributary to the Alamo River, a water of the United States, via the Lilac Drain and Rose Drain, and then ultimately to the Salton Sea.

B. Discharge Points and Receiving Waters

Final effluent is discharged through Discharge Point 001 at Latitude 32° 51' 18" North and Longitude 115° 33' 31" West, to the Dolson Drain. The permitted maximum daily flow limitation is equal to the design capacity of the wastewater treatment plant, which is 2.4 MGD. The discharge consists of disinfected secondary treated wastewater.

C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

Effluent limitations contained in the existing Order R7-2010-0020, as amended by Special Order R7-2011-0018, for discharges from Discharge Point 001 (Monitoring Location EFF-001) and representative monitoring data from the term of the previous Order are as follows in Table F-2:

Table F-2. Historic Effluent Limitations and Monitoring Data

Parameter	Units	Effluent Limitation			Monitoring Data (From November 2010 – February 2015)		
		Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Flow	MGD	2.4	--	--	1.4 ¹	--	2.4 ²
Biochemical Oxygen Demand (BOD) (5 day @ 20 Deg. C)	mg/L	30	45	--	14.1	19.2	--
Removal Efficiency for BOD	%	85	--	--	93 ³	--	--
Total Suspended Solids (TSS)	mg/L	30	45	--	79.5 ⁴	90 ⁵	--
Removal Efficiency for TSS	%	85	--	--	83 ⁶	--	--
pH	s.u.	--	--	6.0 – 9.0 ⁷	--	--	6.7 - 8.2 ⁸
Oil and Grease	mg/L	--	--	25	--	--	31.4 ⁹
<i>Escherichia coli</i> (E. coli)	MPN/100 mL	126 ¹⁰	--	400 ¹¹	21.9	--	120
Fecal Coliform	MPN/100 mL	200 ¹⁰	--	400 ¹²	70	--	1600 ¹³
Copper, Total Recoverable	µg/L	25	--	50	36 ¹⁴	--	36
Silver, Total Recoverable	µg/L	22	--	44	13	--	13

Parameter	Units	Effluent Limitation			Monitoring Data (From November 2010 – February 2015)		
		Average Monthly	Average Weekly	Maximum Daily	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Zinc, Total Recoverable	µg/L	193	--	388	80	--	80
Nitrates as Nitrogen (N)	mg/L	--	--	--	--	--	125
Nitrites as N	mg/L	--	--	--	--	--	1.3
Ammonia as N	mg/L	--	--	--	--	--	18
Total Nitrogen	mg/L	--	--	--	--	--	125
Total Phosphate as Phosphorus (P)	mg/L	--	--	--	--	--	11
Ortho-Phosphate as P	mg/L	--	--	--	--	--	15
Total Dissolved Solids (TDS)	mg/L	--	--	--	--	--	2,586
Temperature	°F	--	--	--	--	--	92
Hardness	mg/L	--	--	--	--	--	852
Acute Toxicity (<i>Ceriodaphnia dubia</i>)	TU _a	15			--	--	0.85
Chronic Toxicity - Reproduction (<i>Ceriodaphnia dubia</i>)	TU _c	15			--	--	1.0
Acute Toxicity (<i>Pimephales promelas</i>)	TU _a	15			--	--	1.0
Chronic Toxicity - Survival (<i>Pimephales promelas</i>)	TU _c	15			--	--	1.0

¹ This value represents the highest average monthly flow value (January, February, and August 2012 and January 2013).

² This value represents the highest reported flow value (January 11, 2011).

³ This value represents the lowest reported value of the minimum percent removal for BOD.

⁴ This value, reported January 2011, represents an effluent limitation violation. The Discharger reported one additional value greater than the average monthly effluent limitation (40 mg/L, February, 2013).

⁵ This value, reported January 12, 2011, represents an effluent limitation violation. The Discharger reported two additional values greater than the average weekly effluent limitation (69 mg/L on January 19, 2011 and 62 mg/L on February 13, 2013).

⁶ This value represents the lowest reported value of the minimum percent removal for TSS (November 2012) and represents an effluent limitation violation. The Discharger reported one additional value below 85% (84%, October 2014).

⁷ This range represents the instantaneous minimum and maximum pH limitations, respectively.

⁸ This figure represents the range of reported pH values.

⁹ This value represents an effluent limitation violation (January 6, 2015).

¹⁰ 30-Day Geometric Mean, based on a minimum of not less than five samples for any 30-day period.

- ¹¹ No sample shall exceed 400 MPN/100 ml.
¹² No more than ten percent of the total samples during any 30-day period shall exceed 400 MPN/100 ml.
¹³ The Discharger reported seven values that are greater than the maximum daily effluent limitation for fecal coliform; these values are presented in Table F-4, below.
¹⁴ This value represents an effluent limitation violation (October 15, 2010).
¹⁵ Section IV.A.1.c of Order R7-2010-0020, as amended by Special Order R7-2011-0018, states “there shall be no acute or chronic toxicity in the treatment plant effluent nor shall the treatment plant effluent cause any acute or chronic toxicity in the receiving water.”

The ROWD described the existing discharge as follows:

Annual Average Effluent Flow – 1.10 MGD

Maximum Daily Effluent Flow – 1.40 MGD

Average Daily Effluent Flow – 1.10 MGD

Table 3 presents the effluent characteristics reported in the ROWD and EPA Form 2A for pH and temperature. The Discharger did not provide data for the other parameters listed in EPA Form 2A; however, the data reported during the permit term are available in CIWQS.

Table F-3. Effluent Characteristics

Parameter	Units	Maximum Daily	Average Daily
pH (Minimum)	S.U.	7.2	--
pH (Maximum)	S.U.	7.4	--
Temperature (Winter)	°F	73	69
Temperature (Summer)	°F	82	77

D. Compliance Summary

A review of the available effluent monitoring data, submitted in the Discharger’s Self-Monitoring Reports, indicate that the Discharger had effluent limitation violations for TSS, oil and grease, copper, and bacterial indicators (fecal coliform), summarized below:

Table F-4. Violations Report Summary

Date of Exceedance	Limit Basis	Parameter	Permit Limitations	Reported Value	Units
10/15/2010	Average Monthly	Copper	25	36	µg/L
1/12/2011	Average Weekly	TSS	45	90	mg/L
1/19/2011	Average Weekly	TSS	45	69	mg/L
January 2011	Average Monthly	TSS	30	79.5	mg/L
7/16/2012	Maximum Daily	Fecal Coliform	400	1600	MPN/100 ml

Date of Exceedance	Limit Basis	Parameter	Permit Limitations	Reported Value	Units
8/1/2012	Maximum Daily	Fecal Coliform	400	1600	MPN/100 ml
9/12/2012	Maximum Daily	Fecal Coliform	400	1600	MPN/100 ml
November 2012	Average Monthly	Minimum TSS percent Removal	≥85	83	% removal
2/6/2013	Maximum Daily	Fecal Coliform	400	900	MPN/100 ml
2/13/2013	Average Weekly	TSS	45	62	mg/L
February 2013	Average Monthly	TSS	30	40	mg/L
5/15/2013	Maximum Daily	Fecal Coliform	400	500	MPN/100 ml
5/22/2013	Maximum Daily	Fecal Coliform	400	1600	MPN/100 ml
6/12/2013	Maximum Daily	Fecal Coliform	400	1600	MPN/100 ml
October 2014	Average Monthly	Minimum TSS percent Removal	≥85	84	% removal
1/6/2015	Maximum Daily	Oil and Grease	25	31.4	mg/L

E. Planned Changes

The Discharger has not indicated that they are planning any changes at the WPCP.

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as an NPDES permit for point source discharges from this facility to surface waters.

B. California Environmental Quality Act (CEQA)

Under Water Code section 13389, this action to adopt an NPDES permit is exempt from the provisions of Chapter 3 of CEQA, (commencing with section 21100) of Division 13 of the Public Resources Code.

C. State and Federal Laws, Regulations, Policies, and Plans

1. **Water Quality Control Plan.** The Water Quality Control Plan for the Colorado River Basin (hereinafter Basin Plan), which was adopted on November 17, 1993, and amended on November 16, 2012, designates beneficial uses, establishes water quality objectives, and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan (including amendments adopted by the Colorado River Basin Water Board to date). In addition, the Basin Plan implements State Water Resources Control Board (State Water Board) Resolution No. 88-63, which established state policy that all waters, with certain exceptions, should be considered

suitable or potentially suitable for municipal or domestic supply. Consistent with this state policy, effluent limitations specified in this Order protect existing and potential beneficial uses of the Dolson Drain, which are described in Table F-5:

Table F-5. Basin Plan Beneficial Uses

Discharge Point	Receiving Water Name	Beneficial Use(s)
001	Dolson Drain (Imperial Valley Drains)	<u>Existing:</u> Fresh Water Replenishment (FRSH); Water Contact Recreation (REC-I) ^{1, 2} ; Non-Contact Water Recreation (REC-II) ¹ ; Warm Freshwater Habitat (WARM); Wildlife Habitat (WILD); and Support of Rare, Threatened, or Endangered Species (RARE) ³ .

¹ Unauthorized use.

² The only REC-1 use that is known to occur is from infrequent fishing activity.

³ Rare, endangered, or threatened wildlife exists in or utilizes some of these waterway(s). If the RARE beneficial use may be affected by a water quality control decision, responsibility for substantiation of the existence of rare, endangered, or threatened species on a case-by case basis is upon the California Department of Fish and Game on its own initiative and/or at the request of the Colorado River Basin Water Board; and such substantiation must be provided within a reasonable time frame as approved by the Colorado River Basin Water Board.

Requirements of this Order implement the Basin Plan.

2. **Thermal Plan.** The State Water Board adopted the Water Quality Control Plan for Control of Temperature in the Coastal and Interstate Waters and Enclosed Bays and Estuaries of California (Thermal Plan) on January 7, 1971, and amended this plan on September 18, 1975. This plan contains temperature objectives for surface waters. The Thermal Plan does not apply these objectives to the Dolson Drain (Imperial Valley Drains), an agricultural drain because agricultural drainage channels do not have a "natural" receiving water temperature.
3. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** U.S. EPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, U.S. EPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on February 13, 2001. These rules contain federal water quality criteria for priority pollutants.
4. **State Implementation Policy.** On March 2, 2000, the State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the U.S. EPA through the NTR and to the priority pollutant objectives established by the Colorado River Basin Water Board in the Basin Plan. The SIP became effective on May 18, 2000, with respect to the priority pollutant criteria promulgated by the U.S. EPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.
5. **Emergency Planning and Community Right to Know Act.** Section 13263.6(a), CWC, requires that "the Colorado River Basin Water Board shall prescribe effluent limitations as part of the WDRs of a POTW for all substances that the most recent toxic chemical

release data reported to the state emergency response commission pursuant to Section 313 of the Emergency Planning and Community Right to Know Act of 1986 (42 U.S.C. Sec. 11023) (EPCRA) indicate as discharged into the POTW, for which the State Water Board or the Colorado River Basin Water Board has established numeric water quality objectives, and has determined that the discharge is or may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to, an excursion above any numeric water quality objective.”

6. **Storm Water Requirements.** U.S. EPA promulgated Federal Regulations for storm water on November 16, 1990 in 40 C.F.R. parts 122, 123, and 124. The NPDES Industrial Storm Water Program regulates storm water discharges from wastewater treatment facilities. Wastewater treatment plants are applicable industries under the storm water program and are obligated to comply with the Federal Regulations.
7. **Endangered Species Act Requirements.** This Order does not authorize any act that results in the taking of a threatened or endangered species or any act that is now prohibited, or becomes prohibited in the future, under either the California Endangered Species Act (Fish and Game Code, §§ 2050 to 2097) or the Federal Endangered Species Act (16 U.S.C. §§ 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the state. The discharger is responsible for meeting all requirements of the applicable Endangered Species Act.
8. **Anti-degradation Policy.** Federal regulation 40 C.F.R. section 131.12 requires that the state water quality standards include an anti-degradation policy consistent with the federal policy. The State Water Board established California’s anti-degradation policy in State Water Board Resolution 68-16 (“Statement of Policy with Respect to Maintaining High Quality of Waters in California”). Resolution 68-16 is deemed to incorporate the federal anti-degradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Colorado River Basin Water Board’s Basin Plan implements, and incorporates by reference, both the State and federal anti-degradation policies. The permitted discharge must be consistent with the anti-degradation provision of 40 C.F.R. section 131.12 and State Water Board Resolution 68-16.
9. **Anti-Backsliding Requirements.** Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 C.F.R. section 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed.

D. Impaired Water Bodies on CWA 303(d) List

The immediate receiving water is the Dolson Drain, which is a part of the Imperial Valley Drains. The 2010 U.S. EPA CWA Section 303(d) list of impaired waters (hereinafter 303(d) List) classifies the Imperial Valley Drains as impaired by chlordane, dieldrin, Dichlorodiphenyltrichloroethane (DDT), endosulfan, Polychlorinated biphenyls (PCBs), toxaphene, and selenium. The Dolson Drain is tributary to the Alamo River via the Lilac Drain and Rose Drain. A sedimentation/siltation Total Maximum Daily Load (TMDL) for the Alamo River was approved by USEPA on June 28, 2002.

The sedimentation/siltation TMDL establishes a numeric target of an annual average in-stream TSS concentration of 200 mg/L. In assigning the waste load allocation (WLA), the TMDL assigned to the City of Imperial of 127.9 tons per year of suspended solids based on a TSS effluent limit equal to double the TSS effluent limits contained in the City’s NPDES

permit. The TSS effluent limits equal to secondary effluent standards as specified in 40 CFR 133 for TSS of 30 mg/L as a monthly average and 45 mg/L as a weekly average are applied to this discharge. If the Discharger were to discharge at its maximum design flow rate of 2.4 MGD every day for a year as a concentration equal to the monthly average limit, the Facility would not exceed the WLA of 127.9 tons per year of sediment.

Calculation of Average Monthly Effluent Limitation (mass-based):

$$\frac{30\text{mg}}{\text{L}} \times 2.4\text{MGD} \times 8.34 = \frac{601\text{lbs}}{\text{day}}$$

Calculation of TSS loading rate:

$$\frac{601\text{lbs}}{\text{day}} \times \frac{1\text{ton}}{2,000\text{lbs}} \times \frac{365\text{days}}{\text{year}} = \frac{110\text{tons}}{\text{year}}$$

The limits established in this Order adequately implement the requirements of the TMDL.

In addition, the 303(d) List classifies the Salton Sea as impaired by arsenic, chlorpyrifos, DDT, enterococcus, nutrients, and salinity. Tributaries to the Salton Sea, including the Alamo River, may be affected by the development of TMDLs for the Salton Sea. No TMDL has been developed to date for the Salton Sea, although a nutrient TMDL is under development for the Salton Sea that may impact the permitted discharges to tributaries to the Salton Sea (i.e., Alamo River).

E. Other Plans, Policies and Regulations

Federal regulations for storm water discharges require specific categories of facilities, which discharge storm water associated with industrial activity (storm water), to obtain NPDES permits and to implement Best Conventional Pollutant Technology (BCT) and Best Available Technology Economically Achievable (BAT) to reduce or eliminate industrial storm water pollution.

The State Water Board adopted Order 2014-0057-DWQ (General Permit No. CAS0000001), specifying WDRs for discharges of storm water associated with industrial activities, excluding construction activities, and requiring submittal of a Notice of Intent by industries to be covered under the Permit. Coverage under the General Permit is required in accordance with I.A.9 of Order 2014-0057-DWQ.

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source dischargers to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in the Code of Federal Regulations: 40 C.F.R. section 122.44(a) requires that permits include applicable technology-based limitations and standards; and 40 C.F.R. section 122.44(d) requires that permits include water quality-based effluent limitations to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) may be established: (1) using U.S. EPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) on an indicator parameter for the pollutant of concern; or (3) using a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in 40 C.F.R. section 122.44(d)(1)(vi).

Effluent and receiving water limitations in this Order are based on the federal CWA, Basin Plan, State Water Board's plans and policies, U.S. EPA guidance and regulations, and best practicable waste treatment technology. While developing effluent limitations and receiving water limitations, monitoring requirements, and special conditions for the draft permit, the following information sources were used.

1. U.S. EPA NPDES Application Forms: California Form 200, U.S. EPA Forms 1 and 2A dated April 3, 2015.
2. Code of Federal Regulations – Title 40.
3. Water Quality Control Plan (Colorado River Basin – Region 7) as amended to date.
4. Colorado River Basin Water Board files related to City of Imperial WPCP NPDES permit CA0104400.

A. Discharge Prohibitions

Effluent and receiving water limitations in this Order are based on the Federal CWA, Basin Plan, State Water Board's plans and policies, U.S. EPA guidance and regulations, and best practicable waste treatment technology.

B. Technology-Based Effluent Limitations

1. Scope and Authority

Section 301(b) of the CWA and implementing U.S. EPA permit regulations at 40 C.F.R. section 122.44 require that permits include conditions meeting applicable technology-based requirements at a minimum, and any more stringent effluent limitations necessary to meet applicable water quality standards. The discharge authorized by this Order must meet minimum federal technology-based requirements based on Secondary Treatment Standards at 40 C.F.R. part 133 and Best Professional Judgment (BPJ) in accordance with 40 C.F.R. section 125.3.

- a. **Secondary Treatment Standards.** Regulations promulgated in 40 C.F.R. section 125.3(a)(1) require technology-based effluent limitations for municipal Dischargers to be placed in NPDES permits based on Secondary Treatment Standards or Equivalent to Secondary Treatment Standards.

The Federal Water Pollution Control Act Amendments of 1972 (PL 92-500) established the minimum performance requirements for POTWs [defined in section 304(d)(1)]. Section 301(b)(1)(B) of that Act requires that such treatment works must, as a minimum, meet effluent limitations based on secondary treatment as defined by the U.S. EPA Administrator.

Based on this statutory requirement, U.S. EPA developed secondary treatment regulations, which are specified in 40 C.F.R. part 133. These technology-based regulations apply to all municipal wastewater treatment plants and identify the minimum level of effluent quality attainable by secondary treatment in terms of biochemical oxygen demand (BOD₅), total suspended solids (TSS), and pH.

2. Applicable Technology-Based Effluent Limitations

- a. This meets the technology based regulations for the minimum level of effluent quality attainable through secondary treatment in terms of BOD₅ and TSS, removal efficiency for BOD₅ and TSS, and pH as summarized in Table F-6, below. Previous Order R7-2010-0020, as amended by Special Order R7-2011-0018, established technology-based effluent limits to meet applicable secondary treatment standards for BOD₅ and TSS, removal efficiency for BOD₅ and TSS, and pH. These effluent

limitations have been carried over from the previous Order. Further, mass-based effluent limitations are based on a design flow rate of 2.4 MGD.

Table F-6. Summary of Technology-based Effluent Limitations

Parameter	Units	Effluent Limitation				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
Flow	MGD	2.4	--	--	--	--
Biochemical Oxygen Demand (BOD) (5 day @ 20 Deg. C)	mg/L	30	45	--	--	--
	lb/day ¹	601	901	--	--	--
BOD ₅ % Removal	%	85	--	--	--	--
Suspended Solids, Total (TSS)	mg/L	30	45	--	--	--
	lb/day ¹	601	901	--	--	--
TSS % Removal	%	85	--	--	--	--
pH	S.U.	--	--	--	6.0 – 9.0	--

¹ Mass-based effluent limitations are based upon a maximum flow of 2.4 MGD.

b. Basis for Limitations:

Table F-7. Basis for Limitations

Parameters	Basis for Limitations
Flow	The design capacity of the treatment plant is 2.4 MGD.
Biochemical Oxygen Demand (BOD) (5 day @ 20 Deg. C)	Discharges to waters that support aquatic life and are dependent on oxygen. Organic matter in the discharge may consume oxygen as it breaks down.
Total Suspended Solids (TSS)	High levels of suspended solids can adversely impact aquatic habitat. Untreated or improperly treated wastewater can contain high amounts of suspended solids.
pH	Hydrogen Ion (pH) is a measure of Hydrogen Ion concentration in the water. A range specified between 6.0 and 9.0 ensures suitability of biological life. This limitation has been adopted in the Basin Plan of the Region.

C. Water Quality-Based Effluent Limitations (WQBELs)

1. Scope and Authority

CWA Section 301(b) and 40 C.F.R. section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards.

Section 122.44(d)(1)(i) of 40 C.F.R. requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, water quality-based effluent limitations (WQBELs) must be established using: (1) U.S. EPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable water quality objectives and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

Table F-8 summarizes the applicable water quality criteria/objectives for priority pollutants reported in detectable concentrations in the effluent or for which effluent limitations existed in the previous Order. The Dolson Drain begins at the point of discharge; therefore, there was no opportunity to collect data from the receiving water from a location upstream of the discharge point. The hardness value used to conduct the Reasonable Potential Analysis (RPA) was 228 mg/L, which is the minimum hardness value of the effluent and thus provides a conservative estimate for effluent limitations. These criteria were used in conducting the RPA for this Order.

Table F-8. Applicable Beneficial Uses and Water Quality Criteria and Objectives

CTR No.	Parameter	Most Stringent Criteria	CTR/NTR Water Quality Criteria				
			Freshwater		Saltwater		Human Health for Consumption of:
			Acute	Chronic	Acute	Chronic	Organisms Only
		µg/L	µg/L	µg/L	µg/L	µg/L	µg/L
6	Copper	18.87	30.43	18.87	N/A		--
7	Lead	9.08	233.12	9.08			Narrative
11	Silver	16.75	16.75	--			--
13	Zinc	240.88	240.88	240.88			--
14	Cyanide	5.20	22.00	5.20			220,000
81	Di-n-Butyl Phthalate	12,000	--	--			12,000

--" No water quality criteria available

"N/A" Not Applicable to the receiving water.

3. Determining the Need for WQBELs

In accordance with section 1.3 of the SIP, the Colorado River Basin Water Board conducted a RPA for each priority pollutant with an applicable criterion or objective to determine if a WQBEL is required in the Order. The Colorado River Basin Water Board analyzed effluent data to determine if a pollutant in a discharge has the reasonable potential to cause or contribute to an excursion above a state water quality standard. For all parameters that have the reasonable potential to cause or contribute to an excursion above a water quality standard, numeric WQBELs are required. The RPA considers criteria from the CTR and NTR, and when applicable, water quality objectives specified in the Basin Plan. To conduct the RPA, the Colorado River Basin Water Board identified the maximum observed effluent concentration (MEC) for each constituent, based on data provided by the Discharger.

Section 1.3 of the SIP provides the procedures for determining reasonable potential to exceed applicable water quality criteria and objectives. The SIP specifies three triggers to complete a RPA:

- Trigger 1 – If the MEC is greater than or equal to the CTR water quality criteria or applicable objective (C), a limit is needed.
- Trigger 2 – If background water quality (B) > C and the pollutant is detected in the effluent, a limit is needed.
- Trigger 3 – If other related information, such as a 303(d) listing for a pollutant, discharge type, compliance history, etc., indicates that a WQBEL is required.

Sufficient effluent and ambient data are needed to conduct a complete RPA. If data are not sufficient, the Discharger will be required to gather the appropriate data for the Colorado River Basin Water Board to conduct the RPA. In accordance with section 1.2 of the SIP, the Colorado River Basin Water Board shall have discretion to consider if any data are inappropriate for use in determining reasonable potential.

The RPA was performed on available priority pollutant monitoring data collected by the Discharger from monthly samples collected during the period from January 2012 through February 2015 (copper, silver, and zinc) and from annual samples collected for priority pollutants during the period from October 2012 through November 2014. Based on the RPA, the discharge demonstrates reasonable potential to cause or contribute to an excursion above a water quality standard for copper.

While cyanide has been detected on a single occasion in the effluent, there is a lack of evidence that would demonstrate a reasonable potential for cyanide to cause or contribute to an exceedances of a water quality standard. Data evaluated in the RPA for priority pollutants reported in detectable concentrations in the effluent are summarized in Table F-9.

Table F-9. Summary of Reasonable Potential Analysis

CTR No.	Priority Pollutant	Applicable Water Quality Criteria (C)	Max. Effluent Concentration (MEC)	Max. Detected Receiving Water Concentration (B)	RPA Result – Effluent Limit Required ?	Reason
		µg/L	µg/L	µg/L		
6	Copper	18.87	21	--	Yes	MEC > C
7	Lead	9.08	0.002	--	No	MEC < C & no B
11	Silver	16.75	< 0.4	--	No	MEC is ND & no B
13	Zinc	240.88	80	--	No	MEC < C & no B
14	Cyanide	5.20	21	--	No	BPJ
81	Di-n-Butyl Phthalate	12,000	7.8	--	No	MEC < C & no B

--" Data are not available. BPJ = Best Professional Judgment

4. WQBEL Calculations

- Final WQBELs are based on monitoring results and following the calculation process outlined in section 1.4 of the SIP. A table providing the calculations for all applicable WQBELs for this Order is provided in Attachment G of this Order.
- WQBELs Calculation Example**

Using copper as an example, the following demonstrates how WQBELs based on a human health criterion were established for Order R7-2015-0030. The process for developing these limits is in accordance with section 1.4 of the SIP. Attachment G summarizes the development and calculation of all WQBELs for this Order using the process described below.

Step 1: For each constituent requiring an effluent limit, identify the applicable water quality criteria or objective. For each criterion determine the effluent concentration allowance (ECA) using the following steady state equation:

$$ECA = C + D(C-B) \quad \text{when } C > B, \text{ and}$$

$$ECA = C \quad \text{when } C \leq B,$$

Where C = The priority pollutant criterion/objective, adjusted if necessary for hardness, pH and translators. In this Order a hardness value from the effluent of 228 mg/L (as CaCO₃) was used for development of hardness-dependent criteria, and a pH of 7 was used for pH-dependent criteria. Upstream receiving water samples were not available.
D = dilution credit, and
B = ambient background concentration

For this Order, dilution was not allowed due to the nature of the receiving water and quantity of the effluent; therefore:

$$ECA = C$$

For copper, the applicable water quality criteria are:

$$ECA_{\text{acute}} = 30.43 \mu\text{g/L}$$

$$ECA_{\text{chronic}} = 18.87 \mu\text{g/L}$$

$$ECA_{\text{human health}} = \text{Not Applicable}$$

Step 2: For each ECA based on aquatic life criterion/objective, determine the long-term average discharge condition (LTA) by multiplying the ECA by a factor (multiplier). The multiplier is a statistically based factor that adjusts the ECA to account for effluent variability. The value of the multiplier varies depending on the coefficient of variation (CV) of the data set and whether it is an acute or chronic criterion/objective. Table 1 of the SIP provides pre-calculated values for the multipliers based on the value of the CV. Equations to develop the multipliers in place of using values in the tables are provided in section 1.4, Step 3 of the SIP and will not be repeated here.

$$LTA_{\text{acute}} = ECA_{\text{acute}} \times \text{Multiplier}_{\text{acute}}$$

$$LTA_{\text{chronic}} = ECA_{\text{chronic}} \times \text{Multiplier}_{\text{chronic}}$$

The CV for the data set must be determined before the multipliers can be selected and will vary depending on the number of samples and the standard deviation of a

data set. If the data set is less than 10 samples, or at least 80% of the samples in the data set are reported as non-detect, the CV shall be set equal to 0.6.

For copper, the following data was used to develop the acute and chronic LTA using Table 1 of the SIP:

No. of Samples Available	CV	Multiplier _{acute}	Multiplier _{chronic}
40	2.17	0.11	0.19

$$LTA_{acute} = 30.43 \mu\text{g/L} \times 0.11 = 3.37 \mu\text{g/L}$$

$$LTA_{chronic} = 18.87 \mu\text{g/L} \times 0.19 = 3.57 \mu\text{g/L}$$

Step 3: Select the most limiting (lowest) of the LTA.

LTA = most limiting of LTA_{acute} or $LTA_{chronic}$

For copper, the most limiting LTA was the LTA_{acute}

$$LTA = 3.37 \mu\text{g/L}$$

Step 4: Calculate the WQBELs by multiplying the LTA by a factor (multiplier). WQBELs are expressed as Average Monthly Effluent Limitations (AMEL) and Maximum Daily Effluent Limitations (MDEL). The multiplier is a statistically based factor that adjusts the LTA for the averaging periods and exceedance frequencies of the criteria/objectives and the effluent limitations. The value of the multiplier varies depending on the probability basis, the coefficient of variation (CV) of the data set, the number of samples (for AMEL) and whether it is monthly or daily limit. Table 2 of the SIP provides pre-calculated values for the multipliers based on the value of the CV and the number of samples. Equations to develop the multipliers in place of using values in the tables are provided in section 1.4, Step 5 of the SIP and will not be repeated here.

$$AMEL_{aquatic\ life} = LTA \times AMEL_{multiplier}$$

$$MDEL_{aquatic\ life} = LTA \times MDEL_{multiplier}$$

AMEL multipliers are based on a 95th percentile occurrence probability, and the MDEL multipliers are based on the 99th percentile occurrence probability. If the number of samples is less than four (4), the default number of samples to be used is four (4).

For copper, the following data was used to develop the AMEL and MDEL for aquatic life using Table 2 of the SIP:

No. of Samples per Month	CV	Multiplier _{MDEL}	Multiplier _{AMEL}
4	2.17	9.03	2.90

$$AMEL_{aquatic\ life} = 3.37 \times 2.90 = 9.8 \mu\text{g/L}$$

$$MDEL_{aquatic\ life} = 3.37 \times 9.03 = 30.4 \mu\text{g/L}$$

Step 5: For the ECA based on human health, set the AMEL equal to the
ECA_{human health}

However, there are no human health water quality criteria for copper; therefore, effluent limitations based on human health criteria could not be calculated.

Step 6: Select the lower of the AMEL and MDEL based on aquatic life and human health as the water-quality based effluent limit for the Order.

AMEL _{aquatic life}	MDEL _{aquatic life}	AMEL _{human health}	MDEL _{human health}
9.8 µg/L	30.4 µg/L	Not Available	Not Available

The lowest (most restrictive) effluent limits for copper are based on aquatic life and were incorporated into this Order.

c. **WQBELs Based on Basin Plan Objectives**

- i. The Basin Plan states that any discharge to a water body with a REC-1 designated use shall not have bacterial densities in excess of the following:
 - (a) ***E. Coli***. The geometric mean bacterial density (based on a minimum of not less than five samples equally spaced over a 30-day period) shall not exceed a Most Probable Number (MPN) of 126 per 100 milliliters, nor shall any sample exceed the maximum allowable bacterial density of a MPN of 400 per 100 milliliters.
 - (b) **Fecal Coliform**. The geometric mean bacterial density (based on a minimum of not less than five samples equally spaced over a 30-day period) shall not exceed a MPN of 200 per 100 milliliters, nor shall more than ten percent of the total samples during any 30-day period exceed a MPN of 400 per 100 milliliters.

Effluent limitations for *E. coli* and fecal coliform are incorporated in this Order. The bacterial indicators of *E. coli* and fecal coliform are used to estimate the presence of pathogens in the wastewater effluent discharged to Discharge Point 001. Effluent limitations for *E. coli* and fecal coliform shall be used as an indicator to determine the effectiveness of the municipal wastewater treatment facilities disinfection system.

- ii. The Basin Plan contains narrative water quality objectives for oil and grease and floating material in surface waters, which state: "All waters shall be free from substances attributable to wastewater of domestic or industrial origin or other discharges which adversely affect beneficial uses not limited to: floating as debris, scum, grease, oil, wax, or other matter that may cause nuisance." In addition, as discussed in section III.D.2 of this Fact Sheet, the anti-degradation provisions of the State Water Board Resolution No. 68-16 state that: "Any activity which produces or may produce a waste or increased volume or concentration of waste and which discharges or proposes to discharge to existing high quality waters will be required to meet waste discharge requirements which will result in the best practicable treatment or control of the discharge necessary to assure that (a) a pollution or nuisance will not occur and (b) the highest water quality consistent with maximum benefit to the people of the State will be maintained." Oil and grease is a pollutant that generally may be found in sanitary waste from households, businesses and industries, and for which POTWs typically are designed to remove. Oil and grease removal is typically achieved during primary treatment. Based on information included in

self-monitoring reports submitted by the Discharger, monthly effluent samples for oil and grease indicate their presence in the effluent (22 detectable concentrations ranging from 1.3 mg/L to 31.4 mg/L). Therefore, the discharge demonstrates a reasonable potential to cause or contribute to an in-stream excursion above the Basin Plan's narrative objective for oil and grease and floating material. This Order carries forward the MDEL for oil and grease, to implement the narrative water quality objective contained in the Basin Plan, which will provide protection of the beneficial uses of the Dolson Drain, and ultimately the Alamo River. The effluent limitation for oil and grease is carried forward from the previous Order and is originally based on the numeric limitation (MDEL) included in the adopted General Order R7-2009-0300, NPDES Permit for Low Threat Discharges to Surface Waters Within the Colorado River Basin Region. Effluent monitoring data provided by the Discharger monthly during the permit term indicate oil and grease has been detected in the effluent.

Table F-10. Summary of Water Quality-based Effluent Limitations

Parameter	Units	Effluent Limitations				
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum
<i>Escherichia Coli</i> (<i>E. coli</i>)	MPN/100 ml	126 ¹	---	400 ²	---	---
Fecal coliform	MPN/100 ml	200 ¹	---	400 ³	---	---
Copper, Total Recoverable	µg/L	9.8	---	30.4	---	---
	lbs/day ⁴	0.20	---	0.61	---	---
Oil and grease, Total	mg/L	---	---	25	---	---
	lbs/day ⁴	---	---	500	---	---

¹ This effluent limitation is expressed as a geometric (or log) mean, based on a minimum of not less than five equally spaced samples collected for any 30-day period.

² This effluent limitation is expressed as a maximum single sample value.

³ No more than ten percent of the total fecal coliform samples collected during any 30-day period shall exceed a MPN of 400 per 100 milliliters.

⁴ The mass-based effluent limitations are based on a design capacity of 2.4 MGD.

- (a) **Total Dissolved Solids:** Discharges of wastes or wastewater shall not increase the total dissolved solids content of receiving waters, unless it can be demonstrated to the satisfaction of the Colorado River Basin Water Board that such an increase in total dissolved solids does not adversely affect beneficial uses of receiving waters.

5. Whole Effluent Toxicity (WET)

Whole effluent toxicity (WET) protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. WET tests measure the degree of response of exposed aquatic test organisms to an effluent. The WET approach allows for protection of the narrative "no toxics in toxic amounts" criterion while implementing numeric criteria for toxicity. There are two types of WET tests: acute and chronic. An acute toxicity test is conducted over a shorter time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth.

The Basin Plan specifies a narrative objective for toxicity, requiring that all waters be maintained free of toxic substances in concentrations that are lethal to or produce other detrimental response on aquatic organisms. Detrimental response includes but is not limited to decreased growth rate, decreased reproductive success of resident or indicator species, and/or significant alterations in population, community ecology, or receiving water biota.

The previous Order contained narrative toxicity language and triggers, and monitoring requirements. The Discharger did not exceed any toxicity triggers during the permit term. The Discharger will conduct toxicity monitoring once per year.

This Order implements the narrative objective for toxicity, requiring there shall be no toxicity in the treatment plant effluent. In addition, the Order establishes thresholds that when exceeded requires the Discharger to conduct accelerated toxicity testing and/or conduct toxicity identification evaluation (TIE) and toxicity reduction evaluation (TRE) studies.

In addition to the Basin Plan requirements, section 4 of the SIP states that a chronic toxicity effluent limitation is required in permits for all discharges that will cause, have the reasonable potential to cause, or contribute to chronic toxicity in receiving waters. Therefore, in accordance with the SIP, this Order requires the Discharger to conduct chronic toxicity testing for discharges to the Dolson Drain.

D. Final Effluent Limitation Considerations

1. Anti-Backsliding Requirements

Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 C.F.R. section 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. The effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order, with the exception of effluent limitations for silver and zinc. The existing Order (R7-2010-0020, as amended by Special Order R7-2011-0018) contained final effluent limitations for silver and zinc. The effluent limitations for silver and zinc are discontinued in this Order based on the consideration of new information (i.e., current discharge monitoring data and reasonable potential analysis). This relaxation of effluent limitations is consistent with the anti-backsliding requirements of the CWA and federal regulations.

2. Anti-degradation Policies

Section 131.12 of the code of federal regulation requires that the state water quality standards include an antidegradation policy consistent with the federal policy. The State Water Board established California's antidegradation policy in State Water Board Resolution No. 68-16. Resolution No. 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution No. 68-16 requires discharges to waters of the State be regulated to achieve the "highest water quality consistent with maximum benefit to the State." It also establishes the intent that where waters of the State are of higher quality than that required by state policies, including Water Quality Control Plans, such higher quality "shall be maintained to the maximum extent possible" unless it is demonstrated that any change in quality will be consistent with maximum benefit to people of the State, will not unreasonably affect beneficial uses, and will not result in water quality less than that described in plans and policies (e.g., violation of any water quality objective). The discharge is also required to meet waste discharge requirements that result in the best practicable treatment or

control necessary to assure that pollution or nuisance will not occur, and that the highest water quality consistent with maximum benefit to the people will be maintained.

The source water for the City of Imperial and the entire Imperial Valley is the Colorado River. Average annual precipitation in the Imperial Valley is insignificant (approximately 2 inches/year). Therefore, the Dolson Drain is an agricultural (effluent)-dominated surface water that also carries discharges from WWTPs; agricultural return flows from approximately 30 Imperial Valley drains that discharge tilewater and tailwater from farmlands. The Dolson Drain discharges to the Alamo River, which in turn discharges to the Salton Sea. Tailwater is irrigation water that does not percolate into the soil, and exits the lower end of the field into a drain. Tailwater tends to erode fields and thus acquire silt and sediments as it crosses and exits a field. Tilewater is water that has percolated through the soil, but is not absorbed by crops. Tilewater flushes salts from the soil. This highly saline water accumulates in tile lines beneath the fields, wherein it is transported to drains by gravity flow or a sump system. Consequently, "background" water quality in the Dolson Drain is difficult to establish for the purpose of conducting a typical anti-degradation analysis. It is likely that the Alamo River has historically contained "background" water from farmland that contains pollutants at concentrations that violate certain Basin Plan water quality objectives for those pollutants, in particular, pesticides, silt/sediment, and selenium. It also contains nutrients (e.g., phosphorus) at concentrations that contribute to the nutrient impairment of the Salton Sea. The agricultural return flows, however, are essentially free of BOD₅ and fecal coliform bacteria and have pH well within the receiving water quality objective of 6.0 to 9.0 pH units.

The discharge from the WPCP contains conventional pollutants (BOD₅, TSS, fecal coliform bacteria, oil and grease, and pH) that are controlled through best practicable control technology currently available (BPT) and best available technology economically achievable (BAT) to prevent exceedances of the receiving water quality objectives for those pollutants and prevent adverse impacts on the REC-I and REC-II beneficial uses of the Dolson Drain. The discharge also contains TDS, but at concentrations significantly below the 4,000 mg/L TDS WQO for the receiving water. The discharge from the WPCP does not contain any of the 303(d) List of impairing pollutants for the receiving water at detectable levels. Therefore, the discharge is not likely to contribute to exceedances of the WQOs for 303(d)-listed pollutants.

Copper, lead, zinc, cyanide, and di-n-butyl phthalate have been measured in the effluent; however, only copper has been detected at concentrations exceeding applicable water quality criteria and is being controlled through a WQBEL derived from water quality criteria established in the CTR. The established WQBELs for copper prevent adverse impacts of the REC I and REC II beneficial uses of the river and ensure compliance with the Basin Plan. Nevertheless, the BOD₅, TSS, bacteria, copper in the discharge are likely to lower water quality in the receiving water (i.e., cause degradation). For conventional pollutants, including BOD₅, TSS and bacteria, this degradation is restricted to pollutants associated with domestic wastewater, is localized and will not result in water quality less than prescribed in the Basin Plan. For toxic pollutants, including copper, this degradation will be not significant once controlled and will not result in water quality less than prescribed in the Basin Plan.

The discharge from the WPCP as permitted herein reflects best practicable treatment and control (BPTC) for the subject wastewater. The control is intended to assure that the discharge does not create a condition of pollution or nuisance and that the highest "background" water quality as defined above will be maintained. The WPCP incorporates:

- a. technology for secondary treated domestic wastewater;
- b. effluent disinfection;
- c. an operation and maintenance manual;
- d. staffing to assure proper operation and maintenance. and
- e. a standby emergency power generator of sufficient size to operate the necessary treatment units during periods of loss of commercial power.

The discharge is necessary to accommodate economic development in the area and essential public services for the City of Imperial, which are an important benefit to the State. Based on the foregoing, the discharge as permitted herein is consistent with Resolution No. 68-16.

3. Stringency of Requirements for Individual Pollutants

This Order contains both technology-based and water quality-based effluent limitations for individual pollutants. The technology-based effluent limitations consist of restrictions on BOD₅, TSS, percent removal, and pH are specified in Federal regulations as discussed in 40 C.F.R. part 133 and the Permit's technology-based pollutant restrictions are no more stringent than required by the CWA. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. Water quality-based effluent limitations have been scientifically derived to implement water quality objectives that protect beneficial uses. Both the beneficial uses and the water quality objectives have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant water quality-based effluent limitations were derived from the CTR, the CTR is the applicable standard pursuant to Title 40, C.F.R. section 131.38. The scientific procedures for calculating the individual water quality-based effluent limitations are based on the CTR-SIP, which was approved by the U.S. EPA on May 18, 2000. All beneficial uses and water quality objectives contained in the Basin Plan were approved under State law and submitted to and approved by U.S. EPA prior to May 30, 2000. Any water quality objectives and beneficial uses submitted to U.S. EPA prior to May 30, 2000, but not approved by U.S. EPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 C.F.R. section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

E. Interim Effluent Limitations – Not Applicable

F. Final Effluent Limitations

Table F-11 below summarizes the proposed effluent limitations for the discharge from the treatment system through Discharge Point 001. Proposed effluent limitations are based on secondary treatment standards, secondary treatment standards, California Toxics Rule, and Colorado River Basin Plan Water Quality Standards.

The previous Order (R7-2010-0020, as amended by Special Order R7-2011-0018) established effluent limitations for the discharge from the WPCP for TSS, BOD₅, BOD and TSS percent removal, and pH based on secondary treatment standards. These effluent limitations have been carried over to the proposed Order.

The previous Order established effluent limitations for *E. coli* and fecal coliform, based on Basin Plan water quality objectives. Effluent limitations for these parameters are carried over to the proposed Order.

Also, this Order carries over a MDEL for oil and grease for discharges from the treatment systems, to implement the narrative water quality objective for aesthetic qualities (i.e., waters free from substances such as debris, scum, grease, and oil), which is based on the limitation included in the adopted General Order R7-2009-0300, NPDES Permit for Low Threat Discharges to Surface Waters Within the Colorado River Basin Region. The Colorado River Basin Water Board determined the measurement of oil and grease helps to ensure that the Discharger is practicing proper operation and maintenance of the Facility and additionally, that the receiving stream and its intended uses are protected. Revised effluent limitations for copper are established in accordance with requirements of the CTR and SIP. This Order discontinues effluent limitations for silver and zinc based on the results of the RPA.

1. Mass-based Effluent Limitations

Title 40 C.F.R. section 122.45(f)(1) requires effluent limitations be expressed in terms of mass, with some exceptions, and section 122.45(f)(2) allows pollutants that are limited in terms of mass to additionally be limited in terms of other units of measurement. This Order includes effluent limitations expressed in terms of mass and concentration. In addition, pursuant to the exceptions to mass limitations provided in section 122.45(f)(1), some effluent limitations are not expressed in terms of mass, such as pH and temperature, and when the applicable standards are expressed in terms of concentration (e.g. CTR criteria and MCLs) and mass limitations are not necessary to protect the beneficial uses of the receiving water.

Mass-based effluent limitations are established using the following formula:

$$\text{Mass (lbs/day)} = \text{flow rate (MGD)} \times 8.34 \times \text{effluent limitation (mg/L)}$$

where: Mass = mass limitation for a pollutant (lbs/day)

Effluent limitation = concentration limit for a pollutant (mg/L)

Flow rate = discharge flow rate (MGD)

2. Final Effluent Limitations

- a. The Discharger shall maintain compliance with the following effluent limitations at Discharge Point 001, with compliance measured at Monitoring Location EFF-001 as described in the MRP.

Table F-11. Summary of Final Effluent Limitations

Parameter	Units	Effluent Limitations					Basis
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum	
Flow ¹	MGD	2.4	---	---	---	---	
Biochemical Oxygen Demand (BOD) (5 day @ 20 Deg. C)	mg/L	30	45	---	---	---	40 CFR 133
	lbs/day ¹	601	901	---	---	---	
Suspended Solids, Total (TSS)	mg/L	30	45	---	---	---	40 CFR 133
	lbs/day ¹	601	901	---	---	---	

Parameter	Units	Effluent Limitations					Basis
		Average Monthly	Average Weekly	Maximum Daily	Instantaneous Minimum	Instantaneous Maximum	
pH	standard units	---	---	---	6.0	9.0	40 CFR 133
Oil and Grease	mg/L	---	---	25	---	---	BPJ
	lbs/day ¹	---	---	500	---	---	
Copper, Total Recoverable	µg/L	9.8	---	30.4	---	---	CTR, SIP
	lbs/day ¹	0.20	---	0.61	---	---	

¹ The mass-based effluent limitations are based on a design capacity of 2.4 MGD.

- b. **Percent Removal:** The average monthly percent removal of BOD 5-day 20°C and TSS shall not be less than 85 percent.
- c. **Toxicity:** There shall be no toxicity in the treatment plant effluent nor shall the treatment plant effluent cause any toxicity in the receiving water, as defined in section V of the MRP. All waters shall be maintained free of toxic substances in concentrations which are toxic to, or which produce detrimental physiological responses in human, plant, animal, or indigenous aquatic life. Compliance with this objective will be determined by use of indicator organisms, analyses of species diversity, population density, growth anomalies, or toxicity tests of appropriate duration or other appropriate methods specified by the Colorado River Basin Water Board.
- d. **Bacteria:** The bacterial concentrations in the wastewater effluent discharged to the Dolson Drain shall not exceed the following concentrations, as measured by the following bacterial indicators:
 - i. **E. Coli.** The geometric mean bacterial density (based on a minimum of not less than five samples equally spaced over a 30-day period) shall not exceed a MPN of 126 per 100 milliliters, nor shall any sample exceed the maximum allowable bacterial density of a MPN of 400 per 100 milliliters.
 - ii. **Fecal Coliform.** The geometric mean bacterial density (based on a minimum of not less than five samples equally spaced over a 30-day period) shall not exceed a MPN of 200 per 100 milliliters, nor shall more than ten percent of the total samples during any 30-day period exceed a MPN of 400 per 100 milliliters.
- e. **Total Dissolved Solids:** Discharges of wastes or wastewater shall not increase the total dissolved solids content of receiving waters, unless it can be demonstrated to the satisfaction of the Colorado River Basin Water Board that such an increase in total dissolved solids does not adversely affect beneficial uses of receiving waters.

G. Land Discharge Specifications – Not Applicable

H. Recycling Specifications – Not Applicable

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

The receiving water limitations in the proposed Order are based upon the water quality objectives contained in the Basin Plan. As such, they are a required part of the proposed Order.

A. Surface Water

The surface water receiving water limitations in the proposed Order are based upon the water quality objectives contained in the Basin Plan and are carried forward from the previous Order. As such, they are a required part of the proposed Order. The receiving water limitations for dissolved oxygen and temperature are as follows:

The discharge shall not cause the concentration of dissolved oxygen in the receiving water to fall below 5.0 mg/L. When the dissolved oxygen in the receiving water is already below 5.0 mg/L, the discharge shall not cause any further depression.

The discharge shall not result in the natural receiving water temperature to be altered, unless it can be demonstrated to the satisfaction of the Colorado River Basin Water Board that such alteration in temperature does not adversely affect beneficial uses.

The discharge shall not result in the normal ambient pH of the receiving water to fall below 6.0 or exceed 9.0 units.

The discharge shall not cause the concentration of total dissolved solids in the Dolson Drain to exceed an annual average concentration of 4,000 mg/L or a maximum daily concentration of 4,500 mg/L.

B. Groundwater – Not Applicable

VI. RATIONALE FOR PROVISIONS

A. Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 C.F.R. section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 C.F.R. section 122.42, are provided in Attachment D. The discharger must comply with all standard provisions and with those additional conditions that are applicable under 40 C.F.R. section 122.42.

Sections 122.41(a)(1) and (b) through (n) of 40 C.F.R. establish conditions that apply to all state-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) of 40 C.F.R. allows the state to omit or modify conditions to impose more stringent requirements. In accordance with 40 C.F.R. section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 C.F.R. sections 122.41(j)(5) and (k)(2) because the enforcement authority under the Water Code is more stringent. In lieu of these conditions, this Order incorporates by reference Water Code section 13387(e).

B. Special Provisions

1. Reopener Provisions

This provision is based on 40 C.F.R. part 123. The Colorado River Basin Water Board may reopen the permit to modify permit conditions and requirements. Causes for modifications include the promulgation of new regulations, modification in sludge use or disposal practices, or adoption of new regulations by the State Water Board or Colorado River Basin Water Board, including revisions to the Basin Plan.

2. Special Studies and Additional Monitoring Requirements

- a. **TRE Work Plan, Toxicity Identification Evaluations, and Toxicity Reduction Evaluations.** This provision is based on the SIP, section 4, Toxicity Control Provisions.
 - b. **Translator Study.** This provision is based on the SIP. This provision allows the Discharger to conduct an optional translator study, based on the SIP at the Discharger's discretion. This provision is based on the need to gather site-specific information in order to apply a different translator from the default translator specified in the CTR and SIP. Without site-specific data, the default translators are used with the CTR criteria.
- 3. Best Management Practices and Pollution Prevention**
- a. **Pollutant Minimization Program.** This provision is based on the requirements of section 2.4.5 of the SIP.
 - b. **Storm Water.** This provision is based on Water Quality Order 2014-0057-DWQ, NPDES General Permit No. CAS000001 for Discharges of Storm Water Associated with Industrial Activities.
- 4. Construction, Operation, and Maintenance Specifications**
- a. **Facility and Treatment Operation.** This provision is based on the requirements of section 122.41(e) and the previous Order.
 - b. **Anti-degradation Analysis and Engineering Report for Significant Expansion.** This requirement is required if the Discharger proposes to significantly upgrade existing treatment systems. The Discharger would be required to evaluate treatment capacity, address mass increases of pollutants discharged, and propose additional units as necessary to enable adequate treatment, while ensuring that any proposed increases in discharges will not violate the State Water Board's anti-degradation policy.
 - c. **Spill Response Plan.** This provision is based on the requirements of section 122.41(e) and the previous Order.
- 5. Special Provisions for Municipal Facilities (POTWs Only)**

The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (General Order) on May 2, 2006. The Monitoring and Reporting Requirements for the General Order were amended by Water Quality Order WQ 2008-0002-EXEC on February 20, 2008. The General Order requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the General Order. The General Order requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the General Order contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. Inasmuch that the Discharger's collection system is part of the system that is subject to this Order, certain standard provisions are applicable as specified in Provisions, section VI.C.5. For instance, the 24-hour reporting requirements in this Order are not included in the General Order. The Discharger must comply with both the General Order and this Order. The Discharger and public agencies that are discharging wastewater into the facility were required to obtain enrollment for regulation under the General Order by December 1, 2006.

- a. **Sludge Disposal Requirements.** Requirements are based on the previous Order and 40 C.F.R. part 503.

- b. **Pretreatment Program Requirements.** Requirements are based on the previous Order and 40 C.F.R. part 403.
- c. **Collection Systems.** Requirements are based on 40 C.F.R. section 122.41.

6. Other Special Provisions

Special Provisions VI.C.6.a and VI.C.6.b are included to ensure the compliance with requirements established in Order R7-2015-0030, and are based on the previous Order, the CWA, U.S. EPA regulations, CWC, and Colorado River Basin Water Board plans and policies.

7. Compliance Schedules

The compliance schedules specify the deliverables and due dates for the TRE Work Plan, Spill Response Plan, and Sludge Disposal Notification and Plan for compliance with the Permit requirements.

VII. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

CWA section 308 and 40 C.F.R. sections 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. Water Code sections 13267 and 13383 also authorize the Colorado River Basin Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The Monitoring and Reporting Program (MRP), Attachment E of this Order establishes monitoring, reporting, and recordkeeping requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this facility.

A. Influent Monitoring

This Order carries forward the treatment plant influent monitoring requirements.

B. Effluent Monitoring

The Discharger is required to conduct monitoring of the permitted discharges in order to evaluate compliance with permit conditions. Monitoring requirements are given in the proposed MRP. This provision requires compliance with the MRP, and is based on 40 C.F.R. sections 122.44(i), 122.62, 122.63 and 124.5. The MRP is a standard requirement in almost all NPDES permits (including the proposed Order) issued by the Colorado River Basin Water Board. In addition to containing definitions of terms, it specifies general sampling/analytical protocols and the requirements of reporting of spills, violations, and routine monitoring data in accordance with NPDES regulations, the CWC, and Colorado River Basin Water Board's policies. The MRP also contains sampling program specific for the Discharger's wastewater treatment facility. It defines the sampling stations and frequency, pollutants to be monitored, and additional reporting requirements. Pollutants to be monitored include all pollutants for which effluent limitations are specified. Further, in accordance with section 1.3 of the SIP, periodic monitoring is required for all priority pollutants defined by the CTR, for which criteria apply and for which no effluent limitations have been established, to evaluate reasonable potential to cause or contribute to an excursion above a water quality standard.

Monitoring for those pollutants expected to be present in the discharge from the Facility, EFF-001, will be required as shown in the proposed MRP and as required by the SIP.

Effluent monitoring requirements are largely unchanged from the previous Order; twice per month monitoring for BOD₅ and TSS is continued at Monitoring Location EFF-001. Further, monitoring five times per month for *E. coli* and fecal coliform, weekly monitoring for temperature, and daily monitoring for pH. Monthly monitoring for copper has been continued to determine compliance with the WQBELs for copper. Monitoring for oil and grease has been maintained monthly to determine compliance with effluent limitations established for oil and grease. Monitoring for nitrite,

nitrate, ammonia nitrogen, total nitrogen, total phosphate, ortho-phosphate, total dissolved solids, and hardness has been reduced from monthly to quarterly since a review of effluent monitoring data indicates low variability.

C. Whole Effluent Toxicity Testing Requirements

Whole effluent toxicity (WET) testing requirements establish monitoring of the effluent to ensure that the receiving water quality is protected from the aggregate toxic effect of a mixture of pollutants in the effluent. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth. This permit requires chronic toxicity testing.

This requirement establishes conditions and protocol by which compliance with the Basin Plan narrative water quality objective for toxicity will be demonstrated. Conditions include required monitoring and evaluation of the effluent for chronic toxicity and numerical values for chronic toxicity evaluation to be used as 'triggers' for initiating accelerated monitoring and toxicity reduction evaluation(s).

This Order modifies the WET testing requirements in that the Order includes a screening phase and a monitoring phase of species testing. Screening is required during the first and third years of the permit term, to determine the most sensitive species that the Discharger will continue to use during the monitoring phase. The Order establishes chronic toxicity testing and monitoring triggers, which when exceeded, initiates accelerated testing, TRE, and TIE procedures. This Order also includes implementation procedures for toxicity caused by ammonia, ionic imbalance, and elevated TDS concentrations.

The WET testing requirements contained in the MRP, section V were developed based on the Draft National Whole Effluent Toxicity Implementation Guidance Under the NPDES Program developed by U.S. EPA (Docket ID. No. OW-2004-0037) and the Test of Significant Toxicity Implementation (EPA 833-R-10-003) and Technical (EPA 833-R10-002) Documents. This is the most current guidance available to the Colorado River Basin Water Board.

The U.S. Environmental Protection Agency (EPA or the Agency) has developed a new statistical approach that assesses the whole effluent toxicity (WET) measurement of wastewater effects on specific test organisms' ability to survive, grow, and reproduce. The new approach is called the Test of Significant Toxicity (TST) and is a statistical method that uses hypothesis testing techniques based on research and peer-reviewed publications. The TST approach examines whether an effluent at the critical concentration (e.g., in-stream waste concentration or IWC, as recommended in EPA's Technical Support Document (TSD) (U.S. EPA 1991) and implemented under EPA's WET National Pollutant Discharge Elimination System (NPDES) permits program) and the control within a WET test differ by an unacceptable amount; i.e., the amount that would have a measured detrimental effect on the ability of aquatic organisms to thrive and survive.

The TST approach explicitly incorporates test power (the ability to correctly classify the effluent as nontoxic) and provides a positive incentive to generate valid, high quality WET data to make informed decisions regarding NPDES WET reasonable potential (RP) and permit compliance determinations. Once the WET test has been conducted, the TST approach can be used to analyze the WET test results to assess whether the effluent discharge is toxic at the critical concentration. The TST approach is designed to be used for a two concentration data analysis of the IWC or a receiving water concentration (RWC) compared to a control concentration. Using the TST approach, permitting authorities will have more confidence when making NPDES determinations as to whether a permittee's effluent discharge is toxic or non-toxic. Use of the TST approach does not result in any changes to EPA's WET test methods; however, a facility might want to modify its future WET tests by increasing the number of replicates over the minimum required (U.S. EPA 1995, 2002a, 2002b, 2002c) by the approved EPA WET test method to

increase test power, which is the probability of declaring an effluent non-toxic if the organism response at the IWC is truly acceptable.

This Order includes a reopener to allow the requirements of this section to be revised pending the issuance of final guidance or policies developed by either the U.S. EPA or State Water Board.

D. Receiving Water Monitoring

1. Surface Water

Surface water monitoring is required to determine compliance with receiving water limitations and to characterize the water quality of the receiving water pursuant to the Basin Plan. Monitoring requirements for the receiving water are largely unchanged from the previous Order. This Order establishes monitoring for nutrients in the downstream receiving water in order to characterize that portion of the Dolson Drain upon the completion of the Salton Sea Nutrient TMDL. Monitoring for nutrients has been reduced from monthly to quarterly based on a review of monitoring data that reveals little variability in the data, in both the upstream and downstream data sets. Additionally, annual monitoring for priority pollutants in the upstream receiving water has been continued, as required in accordance with the SIP.

2. Groundwater – Not Applicable

E. Other Monitoring Requirements

1. Biosolids/Sludge Monitoring

This section establishes monitoring and reporting requirements for the storage, handling and disposal practices of sludge generated from the operation of this Facility. All sludge and or solids generated at the treatment plant will be disposed, treated, or applied to land in accordance with Federal Regulations 40 C.F.R. part 503. The previous Order required sludge monitoring on an annual basis. This monitoring will be carried over from the previous permit.

2. Discharge Monitoring Report-Quality Assurance (DMR-QA) Study Program

Under the authority of section 308 of the CWA (33 U.S.C. § 1318), U.S. EPA requires major permittees under the NPDES Program to participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by U.S. EPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to U.S. EPA's DMR-QA Coordinator and Quality Assurance Manager.

VIII. PUBLIC PARTICIPATION

The Colorado River Basin Water Board has considered the issuance of WDRs that will serve as an NPDES permit for the City of Imperial WPCP. As a step in the WDR adoption process, the

Colorado River Basin Water Board staff has developed tentative WDRs and has encouraged public participation in the WDR adoption process.

A. Notification of Interested Parties

The Colorado River Basin Water Board notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and provided an opportunity to submit written comments and recommendations. Notification was provided through the following the Imperial Valley Press newspaper.

The public had access to the agenda and any changes in dates and locations through the Colorado River Basin Water Board's website at:
<<http://www.waterboards.ca.gov/coloradoriver>>.

B. Written Comments

Interested persons were invited to submit written comments concerning tentative WDRs as provided through the notification process. Comments were due either in person or by mail to the Executive Office at the Colorado River Basin Water Board at 73-720 Fred Waring Drive, Suite 100, Palm Desert, CA 92260.

To be fully responded to by staff and considered by the Colorado River Basin Water Board, the written comments were due at the Colorado River Basin Water Board office by 5:00 p.m. on [INSERT DATE], 2015.

C. Public Hearing

The Colorado River Basin Water Board held a public hearing on the tentative WDRs during its regular Board meeting on the following date and time and at the following location:

Date: September 17, 2015
Time: 9:00 AM
Location: California Regional Water Quality Control Board
Colorado River Basin Region Board Room
73-720 Fred Waring Drive, Suite 100
Palm Desert, CA 92260

Interested persons were invited to attend. At the public hearing, the Colorado River Basin Water Board heard testimony pertinent to the discharge, WDRs, and permit. For accuracy of the record, important testimony was requested in writing.

D. Reconsideration of Waste Discharge Requirements

Any person aggrieved by this action of the Colorado River Basin Water Board may petition the State Water Board to review the action in accordance with Water Code section 13320 and the California Code of Regulations, title 23, sections 2050 and following. The State Water Board must receive the petition by 5:00 p.m., 30 days after the date of this Order, except that if the thirtieth day following the date of this Order falls on a Saturday, Sunday, or state holiday, the petition must be received by the State Water Board by 5:00 p.m. on the next business day. Copies of the law and regulations applicable to filing petitions may be found on the Internet at:

http://www.waterboards.ca.gov/public_notices/petitions/water_quality

or will be provided upon request.

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street

Sacramento, CA 95812-0100

For instructions on how to file a petition for review, see
http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtml

E. Information and Copying

The Report of Waste Discharge, other supporting documents, and comments received are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through Colorado River Basin Water Board by calling (760) 346-7491.

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Colorado River Basin Water Board, reference this facility, and provide a name, address, and phone number.

G. Additional Information

Requests for additional information or questions regarding this order should be directed to Anders Wistrom at (760) 776-8964.

ATTACHMENT G – SUMMARY OF WQBELS CALCULATIONS

The WQBELS developed for this Order are summarized below and were calculated as described in the methodology summarized in Attachment F, Fact Sheet and are contained in section IV.A.1.a of this Order.

Table G-1. Summary of WQBELS Calculations

CTR #	Parameter	Human Health Calculations			Aquatic Life Calculations											Effluent Limitations	
		Organism Only			Freshwater												
		AMEL HH = ECA = C HH only	MDEL/ AMEL multiplier	MDEL HH	ECA acute = C acute	ECA acute multiplier	LTA acute	ECA chronic = C chronic	ECA chronic multiplier	LTA chronic	Lowest LTA	AMEL multiplier 95	AMEL aquatic life	MDEL multiplier 99	MDEL aquatic life	AMEL	MDEL
		µg/L		µg/L	µg/L		µg/L	µg/L		µg/L	µg/L					µg/L	µg/L
6	Copper	--	--	--	30.43	0.11	3.37	18,87	0.19	3.57	3.37	2.90	9.76	9.03	30.43	9.8	30.4